MNemo V2 — User Manual

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Version

This user manual applies to firmware v3.4.2 and higher.

This firmware requires the latest version of Ariane.

> the latest firmware is available here and the upgrade procedure is explained here

If you wish to contribute to this user manual, it can be done on its public repository on GitHub.

The documentation is written in Markdown language.

Change Log Firmware

v3.4.2

A release for the wheel: READY now tells you whether it is on the line, ECO no longer keeps it lit through every shot, and a shot no longer gains a tick as it ends.

- **Is the wheel on the line?** In BASIC and ECO, READY now shows whether the wheel turns: slide the Mnemo a little along the line, and while the wheel turns the screen (in ECO, the disc) blinks between READY's colour and a dimmer, half-strength copy of it. If it stays steady, the wheel is not turning. Nothing is measured, and the wheel's sensors are only flashed for an instant, 100 times a second, about 0.4 mA. In BASIC, a READY colour set with `setcolor` too dark to show READY's black icon at half strength blinks to a lighter shade instead. See Is the wheel on the line?.
- **ECO draws less while it measures.** The wheel's two light sensors draw about 15 mA when lit, and every survey mode kept them lit through the whole of every shot. ECO now flashes them 500 times a second instead, each time just long enough for one reading, so they draw about 2 mA, and they stay dark while the slider is read; the shot is counted exactly as before. See The wheel is flashed, not lit.
- **A shot no longer gains a tick as it ends.** In every survey mode, switching the wheel's sensors off at the end of a shot could count one more tick of the wheel, so the shot could come out one tick, about 2 cm, too long. The switch-off is no longer counted.
- **Calibrated the wheel with AUTO on v3.4.1 or older? Run it again.** The extra tick could get into the calibration's 5 m too, and a calibration that made up for it now makes every shot about 0.5% too short, about 25 cm on 50 m. See Calibrating the distance measurement.
- **The screen saver no longer freezes the Mnemo.** Left idle on USB for an hour or so, the Mnemo could freeze: the screen went black, the battery icon stopped blinking, and the slider did nothing until it was unplugged. It could also happen on battery, on a Mnemo switched on for more than 72 minutes and then left idle on the menu, which then never switched itself off.
- **Smaller fixes**:
  - closing the map in READY brings READY straight back, with its readings: the map's last page used to stay on the screen;

- after a shot or a cancelled shot, READY shows its readings at once rather than a second later;
- BASIC writes in white on a state colour set with `setcolor` too dark for black writing.

> Coming from **v3.4.1**, this update arrives over WiFi as usual. Coming from **v3.4.0** it does too, but the download is still done by v3.4.0's code: if it fails part way, start **OTA UPDATE** again before you switch the Mnemo off (see the note under v3.4.1). Coming from **v3.1.1 or older**, the one-time USB update described in Firmware Update is still needed before over-the-air updates work at all. After updating, run the wheel's AUTO calibration again if you last ran it on v3.4.1 or older.

v3.4.1

A fix release: two features that caused more trouble than they solved are gone, BASIC and the screen saver behave the way you would expect, WiFi passwords full of symbols work, and updates are checked before they are trusted.

- **The double tap no longer opens the map.** Tapping the back of the Mnemo twice used to bring up the current survey's map. In BASIC and ECO, the **LEFT-RIGHT-LEFT-RIGHT** click command still does. In Verbose mode there is no longer a way to see the section you are surveying: its map is under **OPTIONS > HISTORY** once you have ended it. The *Double-tap map display* setting is gone from the web page, and so are the `activatedoubletap` and `deactivatedoubletap` serial commands.
- **The click threshold no longer changes on its own.** Since v2.6.9, four presses in a row that the slider did not register lowered the click threshold and saved it, and the device stopped for two seconds on a **NEW BUTTON THR** message, in the middle of a survey too. The threshold now changes only when you set it: in the **Slider button** card of the web page, or with the `setclickthreshold` serial command. A threshold lowered this way before the update stays as it is until you set it again.
- **The serial click settings refuse out-of-range values.** `setclickthreshold` now ignores values outside 10–90 and `setunclickdelta` outside 2–40, instead of saving them: a typo could leave the slider unusable until the next power-on.
- **BASIC keeps your legs straight.** The IN and OUT click commands (**LEFT-LEFT-LEFT-LEFT** and **RIGHT-RIGHT-RIGHT-RIGHT**) now work only in STANDBY, as this manual has always said: one mistyped command in the middle of a section used to shift every LRUD for the rest of that section by one station. A leg you cancelled, or one that failed to save, no longer leaves its values behind for the next marker or LRUD to write onto the previous

leg. A marker never gives a leg a second number, and the LRUD entered for the next leg no longer lands on the one you just shot. With **LRUD (OFF)** in the settings, one LRUD entry no longer rides along on every later leg, or into the next section. Closing the map no longer leaves its page-turn clicks behind to start the next leg or click command.

- **Leaving BASIC or ECO shows the main menu.** After a BASIC or ECO survey of five minutes or more, the Mnemo used to go straight to the screen saver the moment you left it. The depth gauge, a compass calibration, Test Mode and the Magneto tool (see Tools) now stay on however long you watch them without pressing anything: the depth gauge used to go dark after five minutes, and could then switch the device off. A page left open long enough to doze, such as a map from History, wakes up with the click that closes it.
- **WiFi passwords with symbols are saved whole.** The device's web page used to save a password full of symbols cut short, with no error, so the Mnemo could never join that network. If you added such a network before, add it again. The web page now also says why a network was not saved (no name, a name longer than 32 bytes, a password that is not 8 to 63 characters, a full list), confirms when one was saved or removed, and warns when the WiFi radio's storage did not keep the change. Adding a network with the `addwifinet` serial command before WiFi had been switched on since power-on no longer overwrites the networks already saved.
- **WiFi storage repair**: **OPTIONS > SETTINGS > SYSTEM > WIFI STORAGE > CHECK & REPAIR** checks the WiFi radio's own storage, where the saved networks live, and only if it no longer saves, clears it so that it works again; you then add your networks again. It is meant for older devices whose saved networks vanish, which also keeps them from getting online to update. The `checkwifistorage` and `repairwifistorage` serial commands do the same over USB. See Saved networks are not kept.
- **Safer updates.** A WiFi radio update is now checked after it is written: a write the radio refused used to be reported as a success, which could leave the radio unable to start. **OTA UPDATE** on a low battery now says **Battery too low.** instead of starting a long install. An app download that fails part way no longer installs a broken firmware at the next power-on; this protection comes with v3.4.1, so it covers the updates after this one (see the note below). See Firmware Update.
- **Your settings survive a bad start.** If the memory chip does not answer properly at power-on, the Mnemo now keeps your stored settings instead of resetting the language, the usage timers, the compass and clock types or the compass orientation. While that lasts, the usage times show **?** rather than a wrong number, and the error log shows **13-20**. Errors found while the Mnemo starts up, which the start-up used to wipe, now stay in the error log.
- **Smaller fixes**:

    - the menu no longer freezes after an UNDO that could not read the memory, or after a WiFi radio update that failed;
    - a device still set to the survey mode retired in 2022 now starts a Verbose survey, as its setting has always shown;
    - `setname` accepts 1 to 19 letters, digits, spaces, `-` or `_` and refuses anything else with a message, so the device can no longer be given an empty name, which broke the name of its WiFi access point;
    - **WiFi At Start** stays off on a device that never had it switched on: devices set up on v3.0.0 or older could have it switched on silently by their first v3.0.1 update. If yours did, switch it off at **OPTIONS > WIFI > WIFI ON AT START**;
    - the Chinese power-on prompt draws its last row of text;
    - a full section of 99 legs no longer breaks the map.
- **New in the error log**: `6-1` the motion sensor did not answer a read; `7-0` and `7-1` the motion sensor failed its start-up check; `13-20` the settings could not be read at power-on; `14-0` the WiFi radio did not answer when WiFi started; `21-0` a BNO08x compass did not start (now kept in the log); `23-6` an old over-long device name had overwritten memory; `24-0` a BNO08x compass stopped reporting.

> Coming from **v3.4.0**, this update arrives over WiFi as usual, but its download is still done by v3.4.0's code. If it fails part way, **do not switch the Mnemo off**: start **OTA UPDATE** again until it completes. Switched off with a half-finished download, v3.4.0 would install it at the next power-on, and only a USB update recovers from that. From v3.4.1 on, a failed download is simply discarded. Coming from **v3.1.1 or older**, the one-time USB update described in Firmware Update is still needed before over-the-air updates work at all.

v3.4.0

A battery and data-safety release: a new low-power survey mode, less power drawn everywhere else, nothing lost when the power goes, and a compass calibration that checks it has seen enough of the world before it lets you save.

- **ECO survey mode**: a third survey mode next to BASIC and Verbose, selected from OPTIONS > SETTINGS > SURVEY. It runs exactly like BASIC — same clicks, same wheel, same readings — but shows the survey phase as a single small coloured dot on a black screen instead of filling the display with colour, refreshed half as often. About **27 mA while surveying against BASIC's 55–57 mA**. Your contrast setting is left alone in ECO.

ECO sections are named `E##` instead of `B##` so you can tell them apart later. See ECO Mode.

- **ECO runs the pack flat on purpose**: the low-battery cutoff that stops BASIC and Verbose leaves usable capacity unspent. In ECO the low-battery icon appears in the top-right corner as a *warning* and the survey carries on until the battery is genuinely empty.
- **Recovery after a power cut mid-survey**: if the device loses power in the middle of a section, the next power-on closes that section, reports **Survey recovered**, and the shots already taken are kept. Previously such a section was discarded entirely as soon as a new survey was started after it.
- **The processor rests between jobs**: the main loop now halts the CPU on any pass with nothing to do instead of spinning at full clock for the whole dive. Nothing changes on screen; the battery simply lasts longer.
- **WIFI OFF really powers the radio down**: switching WiFi off used to leave the radio associated and powered, holding the device at about **80 mA instead of its 28 mA idle**. The radio is now taken out of the power budget.
- **WiFi At Start is a surface feature**: the access point started by *WiFi At Start* is held off when the device is switched on underwater, and shut down within about 5 s if the device goes under while it is up. WiFi you switched on yourself from the menu is **never** turned off automatically.
- **Erase memory from the web interface**: a confirmation page that first shows how many surveys and how much memory are stored and offers a DMP or CSV download, then erases only after you type `ERASE` and submit. See Memory management.
- **Storage problems are reported instead of guessed**: a shot the memory refused is no longer counted (you get a warning and can re-shoot the leg), **MEMORY FULL** and **MEMORY ERROR** appear where a write cannot proceed, a download that could not be completed fails instead of arriving as a short but valid-looking file, and a compass calibration that could not be read at boot is no longer silently replaced by a blank one.
- **Shot counts agree everywhere**: the last leg of a section is counted on the map as well as in the history row and the web list, a corrupt record now costs one section instead of every section after it, and the history menu can no longer open a survey that is not listed.
- **Clock sync reports what happened**: synchronising the clock from the browser now confirms success — or says plainly that it failed. Before, a refused sync looked exactly like a successful one. See Date and Time Synchronization.
- **The WIFI_AP menu row is never blank**: while connected to a local network the row now reads **WIFI_AP (SWITCH)**, since selecting it there swaps the device over to its own access point rather than adding one alongside.

- **The update check runs once per power-on**, not on every WiFi connection. Switching WiFi off and on again no longer re-shows the *checking for updates* banner and the *update available* prompt each time. See Firmware Update.
- **The Access Point QR code dismisses itself on later sessions too.** It used to close correctly only on the first connection after power-on; on a reconnect the page was served from the phone's cache, so the device never saw it load and the QR stayed on screen.
- **Memory usage never reads 0% when data is stored.** The figure on the main menu status line — and on the new web erase page — used to round down, so roughly the first 19 shots recorded still showed `0%`.
- **A settings reset no longer changes which hardware the firmware talks to.** *Reset Settings* used to overwrite the stored choice of real-time clock, pressure sensor and compass, which on some devices left the clock or the depth reading not working, with no way to fix it from the device. Those three selections are now left alone, and a pressure-sensor selection naming a sensor that is not fitted is corrected automatically at the next power-on.
- **Compass calibration shows coverage instead of a reading count**: the calibration screen now shows a **COVERAGE** figure and bar, red until the readings pin the compass model down, yellow from 50% when a save becomes possible, green from 90%, the target. Once a save is possible the figure is the best coverage reached so far, and that is the calibration SAVE CAL. stores, so it cannot drop while you reach for the button. Turning the MNemo flat on the spot filled the old progress bar just as well as a proper tumble, yet gave a calibration that could be wrong by tens of degrees while reporting a perfect fit error; that is exactly what the coverage figure detects. See Calibrating the compass.
- **SAVE is refused on low coverage**: pressing SAVE CAL. too early shows **LOW COVERAGE** and keeps the calibration running. Keep nodding and rolling the device until the bar changes colour.
- **The result stays on screen after a save**, with a grade of EXCELLENT, GOOD or FAIR next to the coverage, fit error, field strength and hard-iron offset that were stored, until you press SELECT.
- **CANCEL keeps your previous calibration**: cancelling a calibration used to leave the compass uncalibrated until the next power-on.
- **New calibration instructions**: turn around while nodding and rolling the device by about 45°. Turning it upside down is not needed.
- **The 3-PLANE calibration is removed.** Three rings within 30° of pitch pin the model down poorly, and the free calibration with its coverage check is both quicker and more accurate at any latitude.

> Coming from **v3.3.1**, this update arrives over WiFi as usual. Coming from **v3.1.1 or older**, the one-time USB update described in Firmware Update is still needed before over-the-air updates work at all.

v3.3.1

First public release since v3.1 — it bundles all of the over-the-air work and the multilingual update.

- **Over-the-air (OTA) updates**: the MNemo can now update its firmware over WiFi — both the main firmware and the WiFi radio — with no cable, and it works on battery. Start it from OPTIONS > SETTINGS > SYSTEM > OTA UPDATE, or accept the *update available* prompt shown when WiFi turns on. A one-time USB update is needed first to switch a device over — see Firmware Update.
- **Chinese over the air**: Simplified (□□□□) and Traditional (□□□□) Chinese are included in the over-the-air image — restored after the first slim OTA builds had to leave them out for size. The Chinese menu wording was also reviewed and refined by a native speaker.
- **"What's New" screen**: after every update the device shows the release's headline features once, on the first boot; read them again any time from OPTIONS > SETTINGS > SYSTEM > WHAT'S NEW.
- **WiFi radio compatibility**: the WiFi (NINA) radio firmware and its on-device driver were brought in line for the over-the-air radio update.
- **Robustness**: assorted fixes to the over-the-air update process.

v3.1.0

- **WiFi QR onboarding**: when WiFi is turned on the OLED shows a QR code so a phone can join the network and open the web page without typing the SSID, password or IP address. Works in both Access Point (two QR codes: join then open-page) and local-network (single open-page QR) modes. The QR dismisses itself once the web page actually loads, or with **NEXT** / **SELECT**.
- **Interactive 3D survey view on the web interface**: the 2D `/View` map now has a **View in 3D** button that opens a full-screen 3D viewer. Orbit (left-drag / one-finger), dolly (right-drag), zoom (wheel / pinch) and click a station to re-centre on it. A base-plane grid and a compass-rose East/North/Up widget help with orientation.

- **Depth-coloured survey paths**: the path on the web 2D map is now drawn as per-segment lines coloured by depth (**blue = deepest, red = highest**) instead of a near-invisible flat line. The on-device OLED depth gradient is reversed to match, so the on-device and web views share a single colour convention.
- **Cleaner WiFi signal-strength icons**: the four RSSI icons on the OLED are now visibly distinct (regenerated at 16×16).
- **USB serial reliability**: a long-standing freeze caused by intermittent USB pogo-contact blips while the device was moved is fixed; the serial port stays stable when the device is jostled while connected, and `syncdatetime` no longer hangs if the host disconnects mid-transfer.

v3.0.0

- **Multilingual UI**: on-device menus and messages are now available in English, Spanish, French, German, Simplified Chinese and Traditional Chinese. Language can be changed at any time in OPTIONS > SETTINGS > DEVICE SETTINGS > LANGUAGE.
- **3D map view**: when viewing a survey map (from History or BASIC mode shortcut), press the select button to cycle between 2D map, 3D perspective view and statistics. In 3D mode the device orientation controls the camera angle.
- **WiFi at Start**: new option (OPTIONS > WIFI > WIFI ON AT START) to automatically start the device's WiFi access point on every power-on. The access point turns off automatically after 30 seconds if unused.
- **Web interface network management**: WiFi networks can now be added and removed directly from the device's built-in web page.
- **Menu reorganisation**: Settings is now split into *Device Settings* (display, units, language) and *System* (update, reset). The Extras menu is renamed to Tools and WiFi has its own dedicated submenu under Options.
- **Button auto-calibration**: performing 4 consecutive clicks on either side of the slider automatically calibrates the button press-force threshold.

v2.6.9

- Added check to avoid memory overflow in history build on startup
- Avg and Max depth display values were swapped
- Fixed marker counter initialisation bug

- Added button threshold auto-calibration (4 consecutive clicks on either side)
- Settings are now saved when modified from the web interface

v2.6.8

- Fixed bug that had date reset on each startup since 1/1/25

v2.6.7

- Display the angle difference in imprecision warning
- In Basic mode when LRUD is activated in settings, it will be displayed at the end of each station
- The section name in Verbose mode is checked for uniqueness
- Change distance display format when distance >100ft or m
- Using MNemo given name instead of fixed SSID for Wifi(AP) name
- Text display optimization
- Fixed bug in Map display mode that could lead to device lock.

v2.6.5

- Completed settings available on the webpage of the MNemo. Date/Time can now be synchronized through a web browser.
- Units get displayed for depth length and azimuth readings
- Added moving arrow progress indicator option instead of blinking background. Option available in Setting menu.
- Improvement of UI display, flickering and dirty pixels removed.
- Optional address parameter for command "getdata" allowing to read MNemos memory after reset.
- LRUD for survey OUT is now recorded at WHITE phase of station IN

v2.6.2

- Added icon  to signal the depth variation is bigger than length of line.

v2.6.1

- Support for network scans that find the device on the local network

v2.6.0

- Added 2 new click sequences, active during "Stand By", to set survey direction in BASIC mode. LEFT-LEFT-LEFT-LEFT for IN survey and RIGHT-RIGHT-RIGHT-RIGHT for OUT survey.
- During the ***Stand By*** screen a index of the click combination is displayed.
- In basic mode Naming of section is now incremental (B01,B02 etc), counter gets reset when memory is.
- The name of the section as well as the direction get displayed beneath the shot counter

v2.5.4

- Add a marker with a unique ID to a station in basic mode by using a combination of clicks: LEFT-LEFT-LEFT-RIGHT
- Marker number gets displayed on live and history map
- Live survey map can be displayed in BASIC mode using a combination of clicks: LEFT-RIGHT-LEFT-RIGHT
- You can access the LRUD entry mode in BASIC mode using a combination of clicks: RIGHT-RIGHT-RIGHT-LEFT
- Added statistics page in history section after displaying the map
- In settings there's a new option to use Icons instead of text for BASIC survey indications
- In settings there's a new option to adjust the contrast of the display.
- Menu simplified for water type, unit and survey mode settings. All testing and debugging mode are now regrouped in Test Mode.
- Bold font used in BASIC mode to increase readability.
- BASIC mode input progress bar optimized to show smoother progress.

- Bug Fix: the device would enter sleep mode in Verbose survey, this could lead to DMP corruption (*no data loss*)
- Bug Fix: Update was missing from Menu

v2.4.12

- Battery voltage display has been replaced by a percentage of charge available.
- On the main menu voltage gets displayed in red when < 20% (+/- 40 min of survey left)
- During survey an icon is displayed in BASIC and VERBOSE survey mode indicating low battery (< 20%)
- When battery reaches 0% the device is safely shutdown avoiding memory corruption that could occur if this happened during the survey phase.
- Improvement in the Extra>Test mode, more data displayed and possibility to adjust parameter live with CLI in this mode.

v2.4.9

Feature update:

- Changed doze and sleep time to 5 min and 20 min
- Double Tap sensitivity can now be adjusted
- Support for HW variant with RV-3028-C7 RTC
- Improved accuracy of slider button
- Randomize position of screen saver image

Bug Fixes:

- Fix issue where double tap light threshold was reset
- Pressure sensor change gets now saved after change
- Fixed display pixelization on startup
- When last shot is erased in Verbose mode it gets now also erased from current map
- Allow lower values of click threshold than 20
- Progress bar no longer erases shot number in verbose mode

v2.4.0

- Added interface to save values for LRUD (Left-Right-Up-Down) profile measured or estimated values. This feature is available in Verbose survey mode only for the moment.
- Fixed issue in Double Tap activated map display.
- When in and out azimuth difference is bigger than 5deg an icon will indicate it.
- Added display of Wi-Fi connection quality on main screen. Wi-Fi auto turn off after 5 min of inactivity.
- Added Setting page to webserver allowing to adjust most MNemo settings through the Wi-Fi interface.
- Text display uses now anti antialiased fonts.

v2.3.0

- Added Warning icon on display when magnetic interference detected
- Record temperature and time at each station.
- Increased size of font in the menu of the Verbose survey mode
- Added CSV file download to the webserver
- Added individual Section CSV download link on the main webpage
- Check for valid date at start
- When wheel mode (WM) is different than 0, reverse direction functionality is disabled
- Fixed bug in display when depth >100

v2.2.18

- Added MN logo to flash screen instead of white background
- Added reset default settings to menu.
- When compass calibration is finished progress bar turns from red to yellow
- Added command to adjust pic switch on threshold. New firmware for PIC

v2.2.16

- Replaced blinking led in Doze mode by blinking screen

- Changed OLED refresh rate

v2.2.14

- Added white blinking led in Doze Mode
- Fixed bug with surface pressure setting getting corrupted

v2.2.13

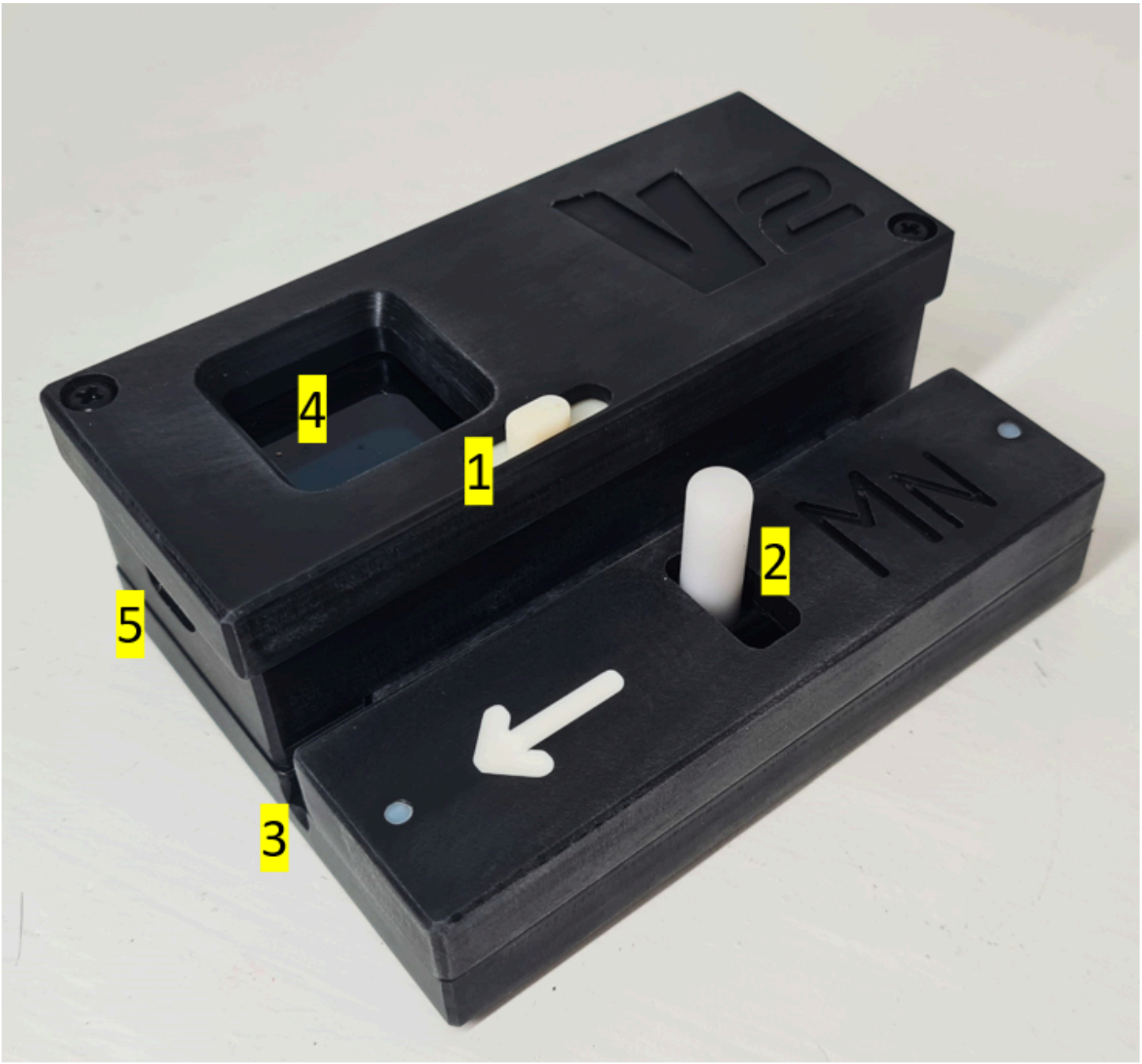
- Added CLI command to activate/deactivate the requirement to move the cursor right to turn on the MNemo: activatesafeon / deactivatesafeon
- Added message indicating that parameters have been updated with the firmware

v2.2.12

- Changed BASIC mode display, added display of station number and color change on whole screen.
- Added setting to deactivate double tap map display. Accessible through CLI: activatedoubletap/deactivatedoubletap
- Added setting to control length of click in BASIC mode. Accessible through CLI: setBMclickfactor factor (factor between 20% and 150%).
- Support for BNO085 as secondary compass.
- Fixed bug when SSID or wifi password contain whitespace character.
- Fixed bug in month settings

Device description

Thank you for your interest in the MNemo! We will start with a short description of the components of the device



1 - Slider Control

It's through this slider that you're gonna interact with the MNemo.

- Moving the slider **Left** is the equivalent of an *Enter* or *Select* order.

- Moving the slider **Right** is the equivalent of a *Next* or *Scroll* order.

2 - Pressure Plate lever

This is the part you will have to pull back to open the device an lock the cave line in it.

> Tip: When pulling the plate back apply the pressure at the bottom of the lever

3 - Line Slot

This is where the cave line will slide though the device after opening and releasing the pressure plate lever.

> The device will fit line up to 6mm diameter

4 - Display

The device has an OLED graphical color display. The text, icon and background colors will guide you through the survey process.

5 - USB Port

This is the port used to connect your device to your computer to adjust settings or download data

> The recommended way to download data is through wifi either by connecting your device to the local network or activating it as Wifi Access Point

It is also the port used for charging the device. Any 5V USB charger or computer USB port can be used to recharge the MNemo.

Switching Mnemo ON

There's no **ON/OFF** switch on the device. You'll switch the device ON using the slider control described in the previous section.

- Move the slider control to the **left** and hold it there until a led turns on, on the right side of the screen. At that moment move the slider to the right and the device will turn on displaying info about the firmware and the serial number of the device.

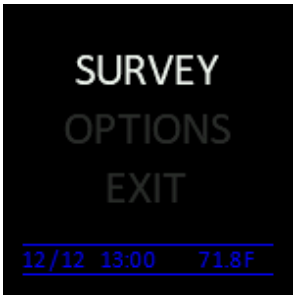
> Tip: If you can’t see the led just count to 6 before moving the slider to the right.

- When this appears, it means MNemo has booted properly, and all the sensors are working. Move again the slider to the left and you’ll be in the main menu.

> Tip: You can deactivate the requirement to move the cursor to the right with the *SwitchONSafety* setting.

Menu Navigation

Each menu will have an active or selected item that will be surrounded by **>....<** or **highlighted**.





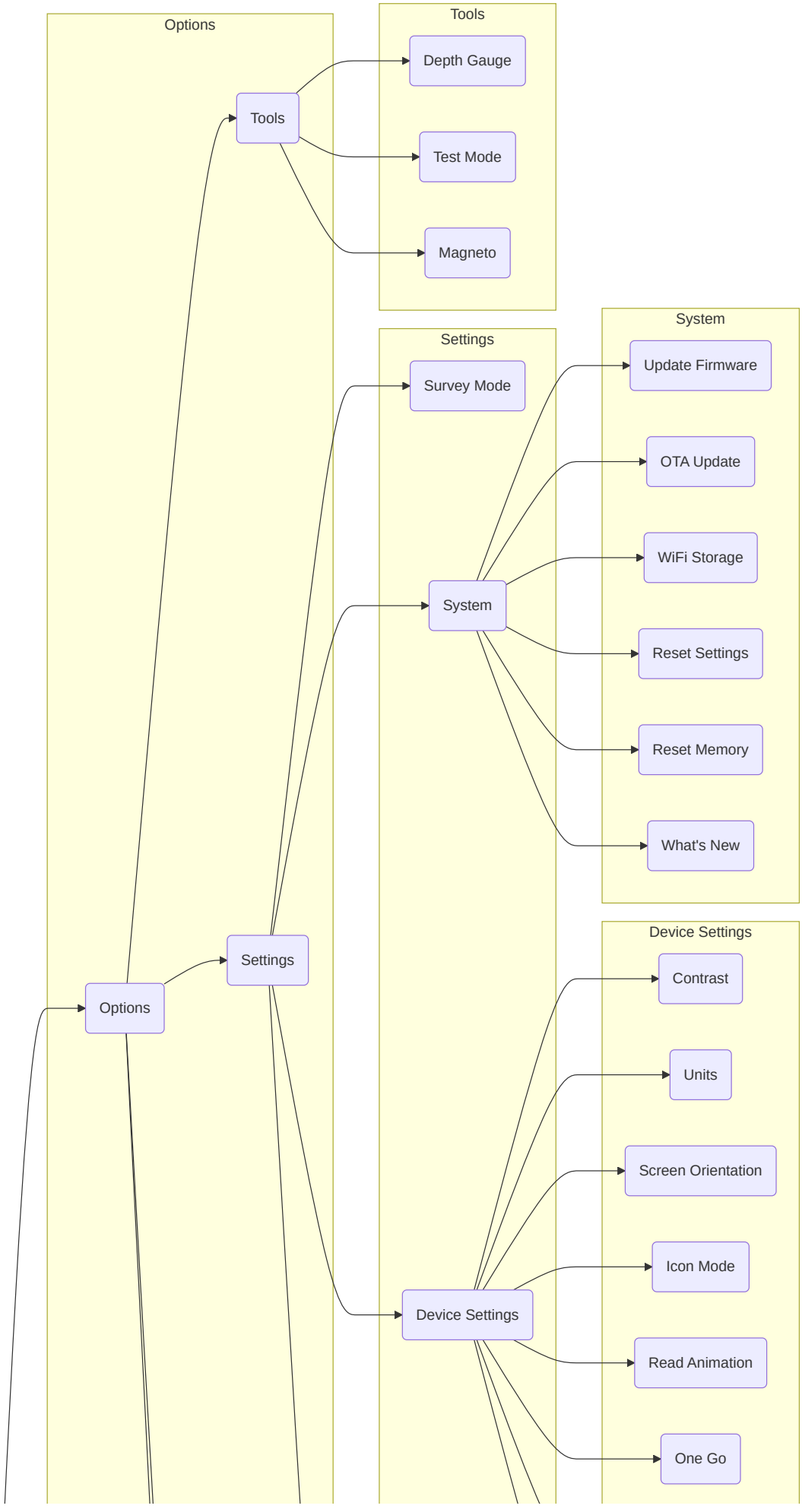
Moving the slider to the **right** is shifting the selected item to the next one (Either on the right or beneath).

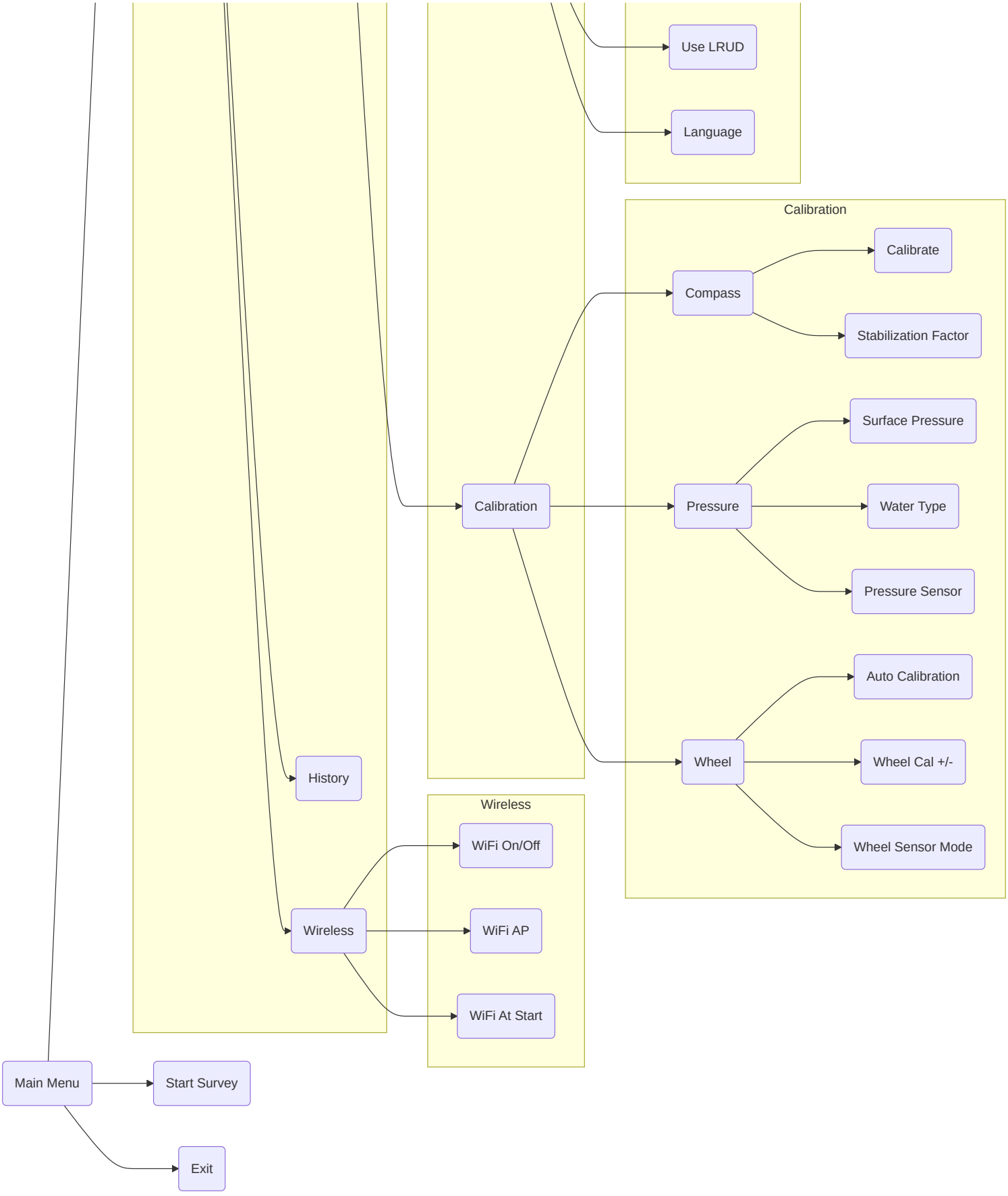
Moving the slider to the **left** is equivalent to the “Enter” key of your keyboard. It selects the current item or starts the action highlighted or selected.

Menus Overview

This document provides a visual representation of the MNemo menu structure.

Main Menu Structure


Options
Tools
Depth Gauge
Test Mode
Magneto
Settings
Survey Mode
System
Update Firmware
OTA Update
WiFi Storage
Reset Settings
Reset Memory
What's New
Device Settings
Contrast
Units
Screen Orientation
Icon Mode
Read Animation
One Go
Options
Tools
Settings
System
Device Settings


Use LRUD
Language
Calibration
Calibrate
Compass
Stabilization Factor
Surface Pressure
Calibration
Pressure
Water Type
Pressure Sensor
Auto Calibration
Wheel
Wheel Cal +/-
Wheel Sensor Mode
History
Wireless
WiFi On/Off
Wireless
WiFi AP
WiFi At Start
Main Menu
Start Survey
Exit

Main Menu

You will notice that the bottom line of the menu is displaying a stream of information:

Display	Description
26/03 11:26 27.6ºC	*Day/Month Hour:Minute Water Temperature*
84% (FW)(m) M:0%	*Battery Charge % - Type of water - Unit - Memory usage*

Both the date/time itself and its display format (Day/Month or Month/Day) can be set from the device's built-in web page over WiFi. See Date and Time Synchronization for the full procedure and details on the consequences of skipping it.

> Tip: If the date is not set or set incorrectly it will be adjusted to a default date. You'll get a **DATE RESET** message when you turn the Mnemo on.

Calibrating the unit

Before starting a survey MNemo’s sensors need to be calibrated and certain parameters need to be adjusted

- Setting the units used by MNemo (Meter or feet)
- Selecting the type of water you are diving in (Fresh or Salt)
- Adjusting the surface pressure
- Adjusting the line measurement factor
- Calibrating the Compass

> **All the settings and calibration are kept in memory when you turn MNemo off.**
> *This is a major difference with Mnemo v1 that required calibration of the compass for each session.*

Setting the Unit System

You can select either the metric system (using meter and ºC) or the imperial system (using feet and ºF) to be displayed on the MNemo. In the menu navigate to :

OPTIONS > SETTINGS > DEVICE SETTINGS > UNITS *(m or ft)* to select the active unit system.

> The unit system used on the device can be different from the one used in your mapping software

Calibrating the distance measurement

Manual Calibration

In the menu navigate to :

OPTIONS > SETTINGS > SENSORS > DISTANCE

and adjust the factor by small changes using the + and - menu entries. After each change take a length measurement of a line with a known length and compare the result. If the measurement is too short increase the factor otherwise decrease it.

> This is not the recommended way to calibrate the length. The Auto mode described below is recommended.

Automatic Calibration

- Put in place a line with markers separated by 5m precisely if the unit is set to METER, or 15 ft if set to IMPERIAL.
- Select **OPTIONS > SETTINGS > SENSORS > DISTANCE > AUTO**
- Take a measure between those two markers as if they were two stations(tie-offs) in a cave.
  > The Standard survey process is explained here
- Once done you'll get a message Adjustment : X , X being the adjustment factor *(How much the factor needs to be changed to have a good measure)*.
- Repeat that operation until you get an adjustment factor of **0** or **1**.

> **Calibrated on v3.4.1 or older? Run the Automatic Calibration again** *(v3.4.2+)*. Before v3.4.2, switching the wheel off at the end of a measurement could add one tick to it, about 2 cm, the calibration's 5 m included. A calibration that made up for that tick makes every shot about 0.5% too short once the tick is gone.

> Tip: If at any point during a survey dive you realize that the line measurement is not working anymore, change the parameter WM to the value 1 or 2 instead of 0. Try both

> setting to find the one that is still working.

Calibrating the compass

This is by far the most important calibration of the MNemo.

- To have a precise and correct calibration it is CRUCIAL to be far away from any magnetic or electromagnetic disturbance, in particular : Scooters, steel tanks, compasses, dive computers, dive light canisters.
- Avoid also: Electric line, cars, any electric motor, metallic structure, reinforced concrete walls etc …

From the main menu select

OPTIONS > SETTINGS > SENSORS > COMPASS

This opens the compass menu with **CALIBRATE** and the stabilization factor setting.

How the calibration works

The compass measures the Earth's magnetic field, but it also sees the small permanent magnetism of the MNemo itself and the way its metal parts bend the field. The calibration measures both so they can be subtracted from every reading you take afterwards.

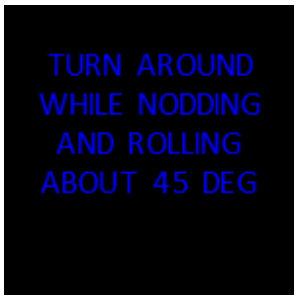
To do that the device needs to see the Earth's field **from all around**: it collects readings while you move it, and fits them to a model of what a fixed field looks like when the device turns through every orientation. The MNemo checks that the readings it collected really do pin that model down, and shows the result as a **COVERAGE** figure. This check is what makes the difference between a calibration you can trust and one that only looks good:

> Turning the MNemo flat on the spot, without tilting it, draws a single circle of readings. That circle fits many different models equally well, so the device would pick one at random. Such a calibration can be wrong by tens of degrees while still reporting an excellent fit error, which is why the MNemo refuses to save it: its coverage stays near 0%.

Running the calibration

Select **CALIBRATE** from the compass menu. An instruction screen is shown for a few seconds:

TURN AROUND / WHILE NODDING / AND ROLLING / ABOUT 45 DEG

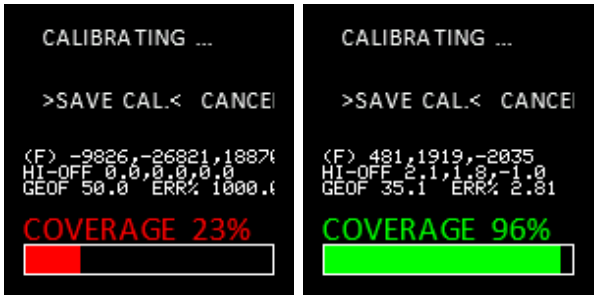


Then the calibration screen appears, with **SAVE CAL.** and **CANCEL** on the menu line.

Hold the MNemo in front of you and turn slowly on the spot, a full turn in about 15 seconds. **While turning, keep nodding the device**, nose up and nose down by about 45°, **and rolling it** from side to side by about 45°, as if the nose of the device were drawing loops. Turning it upside down is not necessary. Keep going until the bar at the bottom of the screen turns **green**, which typically takes 20 to 40 seconds.

The screen shows, from top to bottom:

- The raw magnetometer readings.
- Once enough readings are stored, the hard-iron offset (**HI-OFF**), the geomagnetic field strength (**GEOF**) and the fit error (**ERR%**), recomputed twice a second as you move.
- A count such as **120/192** while the device is still collecting the readings it needs for its first computation, then the **COVERAGE** figure.
- The bar at the bottom. Its colour, and that of the coverage figure, tells you where you stand: **red** while the calibration cannot be saved yet, **yellow** from 50% on, when a save is possible but the calibration is only fair, and **green from 90%**, which is the target.



Once a save is possible, the figure is the **best coverage reached so far**, and that is the calibration SAVE CAL. stores: the seconds you spend reaching for the button, moving less, cannot lose it. Nothing on this screen moves while the device is held still, either. The device

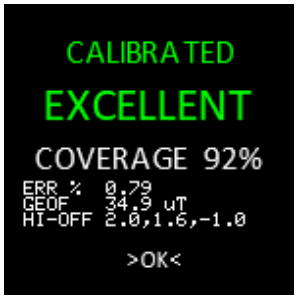
only keeps readings that differ from the ones it already has, so the count and the coverage progress with your movement, not with time.

Aim for **90% or more**: that is what a proper nod-and-roll tumble reaches, usually within a minute, and the figure will not climb much beyond it, so there is no need to chase 100%. The higher the coverage, the less the calibration depends on the quality of the individual readings.

Saving or cancelling

- Select **SAVE CAL.** once the bar is green, or yellow if you are pressed for time. The calibration is stored and kept across power cycles.
- If you select **SAVE CAL.** too early, **LOW COVERAGE** is displayed in red for a few seconds and the calibration carries on: keep moving the device and try again when the bar has changed colour.
- **CANCEL** leaves the calibration screen and keeps the calibration the device had before.

After a save the result stays on screen until you press **SELECT**:



- A grade: **EXCELLENT** (coverage 90% or more), **GOOD** (70% or more) or **FAIR** (below that, or a fit error above 2%). A FAIR calibration works, but it is worth redoing when you can, with more tilt or away from whatever disturbed the field.
- The coverage, the fit error (**ERR%**), the field strength (**GEOF**) and the hard-iron offset (**HI-OFF**) of the calibration that was stored, in case you want to note them down.

Understanding the calibration results

The relevant information is the amount after ERR% which indicates how close the calibration brought the compass to the theoretical model (uniform magnetic field in any orientation). This value should be smaller than 1%.

GEOF is the approximation of the magnetic field at that location. The compass of the Mnemo is not calibrated to give precise absolute magnetic measurements, nevertheless that gives you an indication of the strength of the magnetic field at your location and should be close to the theorical value (NCEI Geomagnetic Calculators)

COVERAGE tells you how well the readings you collected pin the model down. 50% is the minimum the device accepts; 90% or more is what a proper tumble reaches and what the grade calls EXCELLENT. Note that a low ERR% on its own is not a sign of a good calibration: it is the coverage that guarantees the readings could not have fitted a wrong model just as well.

> After a firmware update is generally recommended do redo a calibration of the compass but it is important to turn the device completely off before doing the calibration or you'll get incoherent measurements

> **Important:** A precise and correct compass calibration is essential for obtaining accurate survey data. The calibration normally takes well under a minute. If the bar does not turn green after a couple of minutes, you are most likely turning the device without tilting it enough, or standing close to a magnetic disturbance: vary the tilt more, or move away and start again.

Survey Operations

The Mnemo has three survey modes. You can select which one you prefer to use in

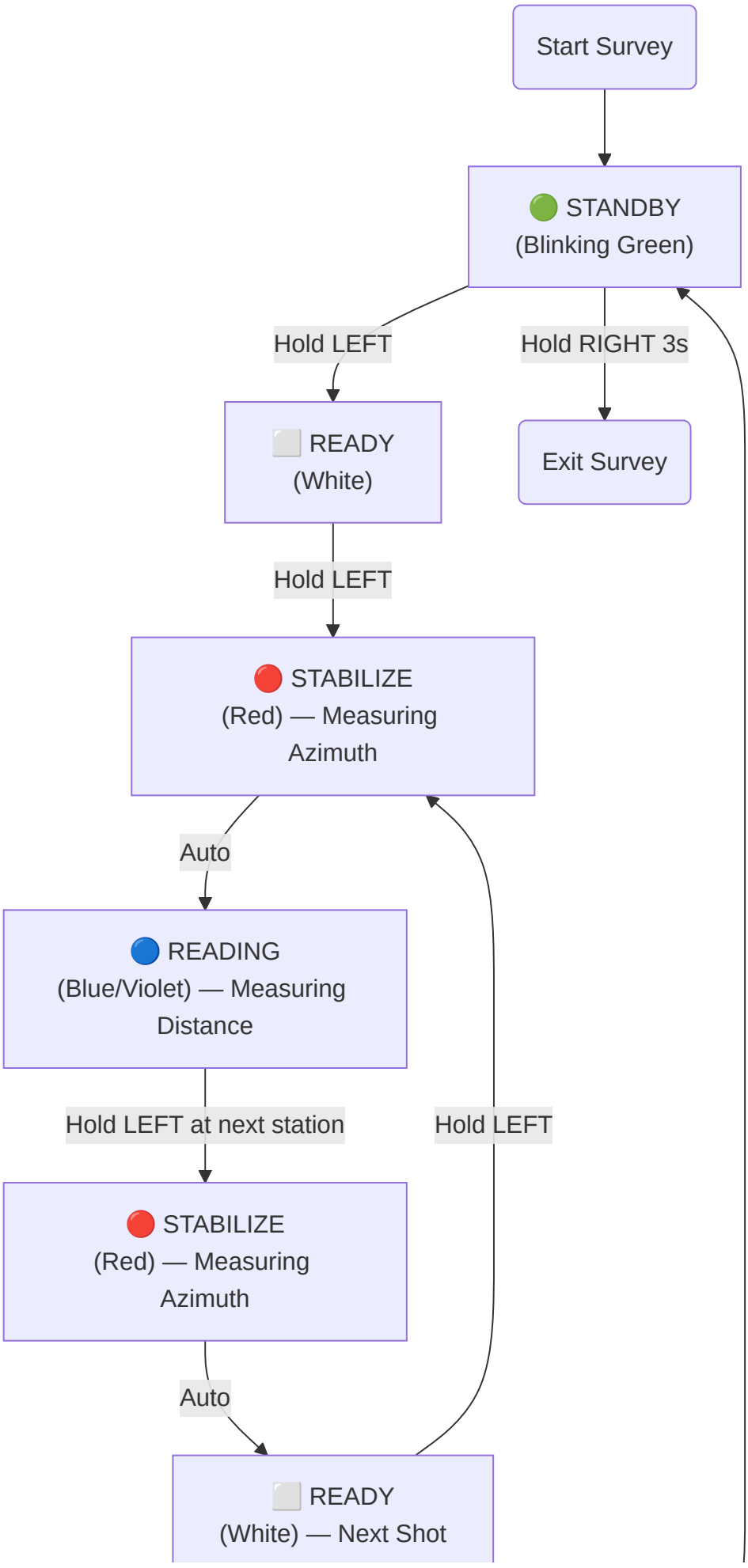
OPTIONS > SETTINGS > SURVEY (BASIC *or* Verbose *or* ECO)

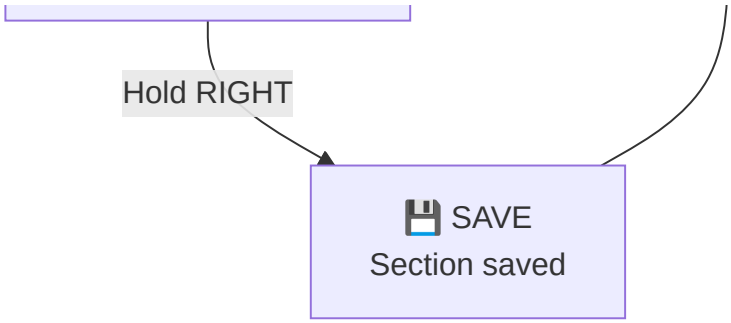
Selecting that menu row cycles through the modes — **BASIC → Verbose → ECO → BASIC** — and the row always shows the one currently selected.

- **BASIC** : This is the recommended survey mode. It does not require from you to read the display, just recognise the background color.
- **ECO** *(new in v3.4.0)* : The same survey as BASIC, drawn as a single small coloured dot on a black screen. It uses about **half the power**, and it deliberately keeps surveying past the low-battery cutoff instead of shutting down — for long dives.
- **Verbose** *(legacy - it will be removed in future versions of the firmware)* This one is easier to start with and offers some options not available in BASIC mode. It requires to have the ability to read the text on the LCD.

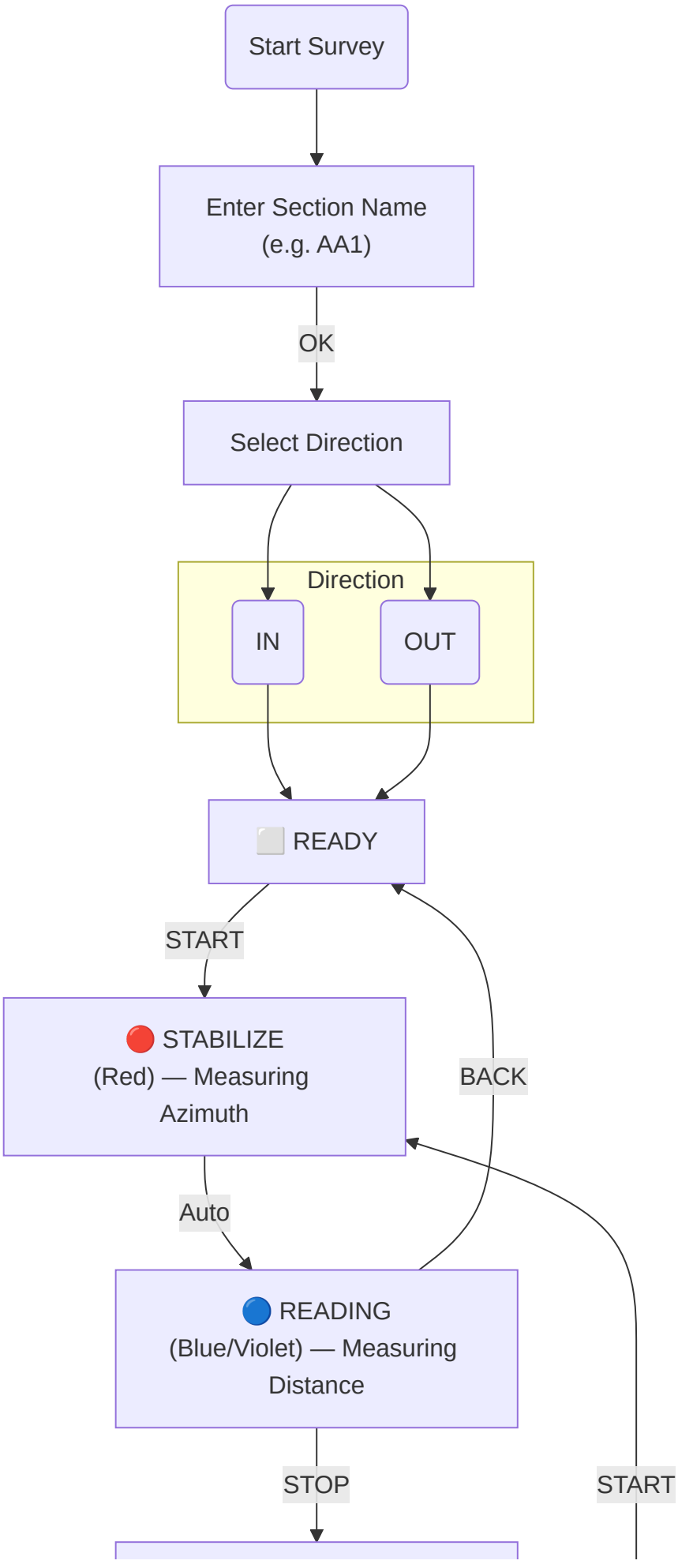
Survey Flow

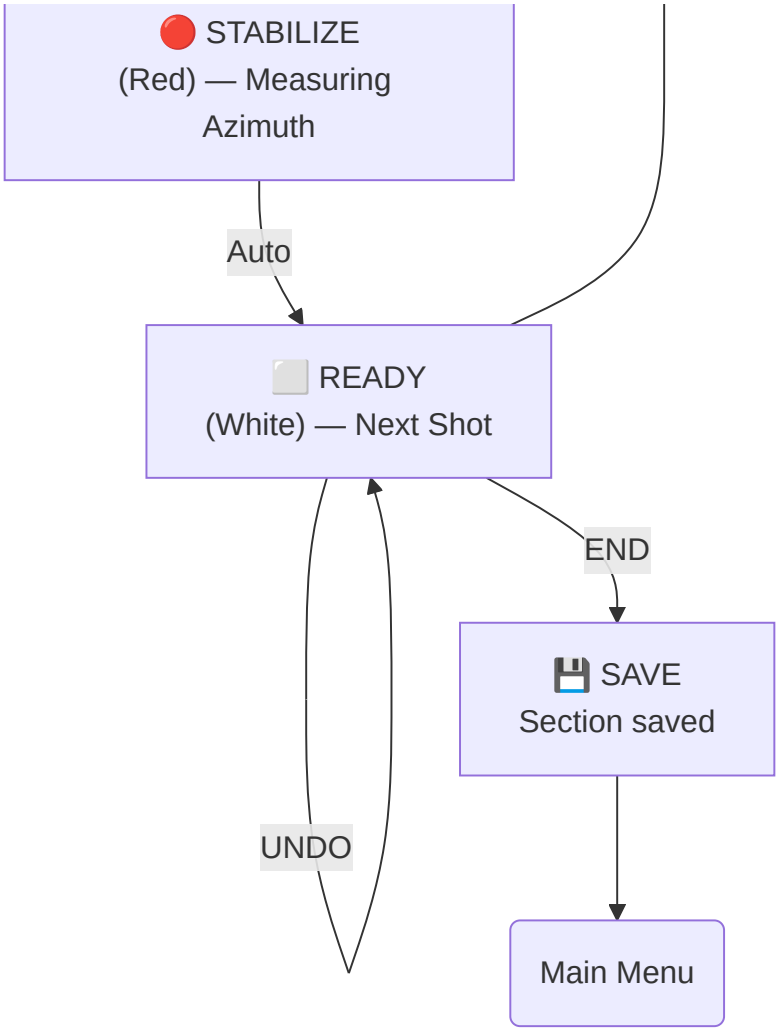
BASIC Mode


Start Survey
🟢 STANDBY
(Blinking Green)
Hold LEFT
Hold RIGHT 3s
⬜ READY
(White)
Exit Survey
Hold LEFT
🔴 STABILIZE
(Red) — Measuring
Azimuth
Auto
🔵 READING
(Blue/Violet) — Measuring
Distance
Hold LEFT at next station
Hold LEFT
🔴 STABILIZE
(Red) — Measuring
Azimuth
Auto
⬜ READY
(White) — Next Shot



Verbose Mode


Start Survey
Enter Section Name
(e.g. AA1)
OK
Select Direction
Direction
IN
OUT
⬜ READY
START
🔴 STABILIZE
(Red) — Measuring
Azimuth
BACK
Auto
🔵 READING
(Blue/Violet) — Measuring
Distance
STOP
START



Color Reference

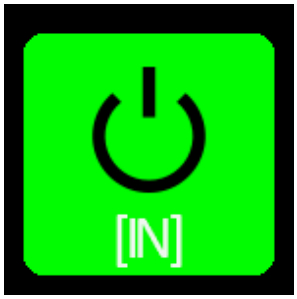
Color	Phase	Description
🟢 Green (blinking)	Standby	Waiting to start a section
⬜ White	Ready	Ready to begin a shot. In BASIC and ECO it blinks to a dimmer white while the wheel turns, so you can check the wheel is on the line *(v3.4.2+)*
🔴 Red	Stabilize	Compass is measuring the azimuth — keep the device still
🔵 Blue / Violet	Reading	Wheel is measuring the distance — move along the line

BASIC Mode (Recommended)

From the main menu select **SURVEY**

> NOTE: The KNOT mode of Mnemo v1 has been removed

You'll notice a flashing ***green*** screen displaying STANDBY or an icon as well as the direction selected for the survey (IN is default but it can be changed using a click command)



> You can select in **OPTIONS > SETTINGS > UI(ICON or TEXT)** if you wish to have the text or the icon displayed in each phase of the survey

From now on you don´t need to be able to read what's displayed on the screen. The color alone will give you information on which phase of the survey you are in.

> In BASIC mode the slider button doesn't work by impulses, you'll have to hold the slider during a certain amount of time (more or less 2s) to give and validate a SELECT or NEXT command. A red progress bar on the top screen will show the progress, the command only gets validated once the progress bar is complete.
>
> 

- From STANDBY mode, to start a section hold the slider left the screen turns WHITE, this indicates READY mode.



> By default, in any BASIC survey mode all surveys are recorded IN and are named “BAS”. You can then later sort out the data on your favorite survey software.

- Clip the device on the line at the beginning of the first shot.
- Slide it a little along the line: if the screen blinks, the wheel is on the line *(v3.4.2+)*. See Is the wheel on the line?
- Hold the slider left the background turns RED, you are in STABILIZE mode (same as in Verbose survey mode).



- Once stabilized, the device goes in READING mode, you can now move along the line, you’ll recognize the flickering BLUE/VIOLET square you already had in Verbose survey mode indicating the wheel is measuring the line



- Once you reach the end of the shot hold the slider left again, the background turns RED, you are in STABILIZE mode.



- Once stabilized the background becomes white again, you are back in READY mode.



- You can repeat the operation for the following shots.

> 
>
> The curve icon you can see in the last picture indicates that the measurement of the azimuth at the start and the end of the shot has a difference of more than 5 degrees. The exact difference is displayed over the curve (In the example we have 7deg of difference)
>
> 
>
> You might also find this icon which signals the change in depth is bigger than the length of the line. This usually happens on very steep shots.

Once you have finished the section, when you are in READY mode, to save the section move the cursor to the right. This will save the data and put the device back in STANDBY mode (Blinking GREEN).

To exit BASIC survey mode, from the STANDBY mode, hold the cursor to the right during 3s.

> **On a long dive, see ECO Mode** *(v3.4.0+)* — the same survey, the same clicks, drawn as a single small coloured dot on a black screen for about half the battery draw.

Is the wheel on the line? *(v3.4.2+)*

You can check that the line runs through the wheel before you start a shot. In READY, slide the MNemo a few centimetres along the line, slowly or briskly:

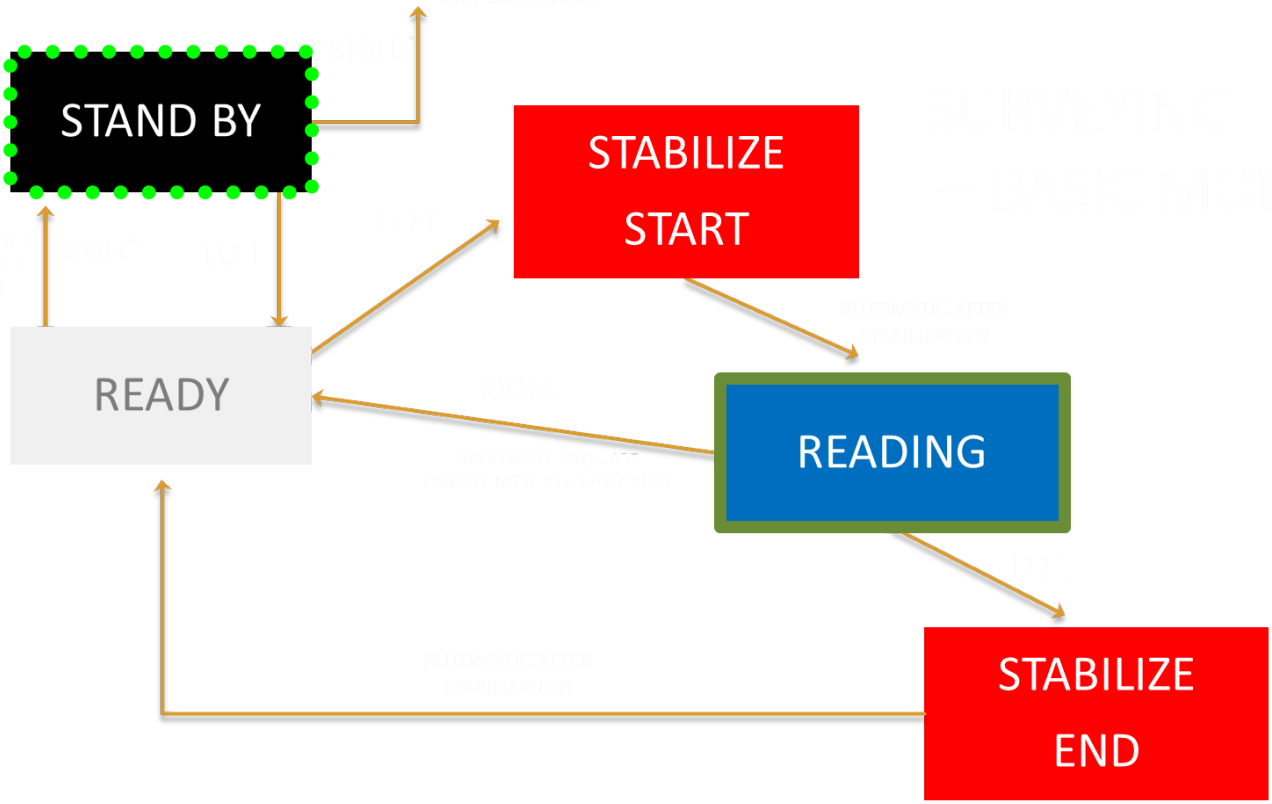
- **The screen blinks** between white and a dimmer, half-strength white: the wheel is turning, so the line is in it.
- **The screen stays plain white**: the wheel is not turning. Check that the line sits in the wheel and is tight enough (see Tips and Tricks).

About a second after you stop, the screen is plain white again.

Nothing is measured while you check: the screen keeps showing your last shot, and the next shot is measured only once you start it. Nor does the check cost much battery: in READY the wheel's sensors are never left on; they are only flashed for an instant, 100 times a second, about 0.4 mA in all.

In ECO the small disc blinks the same way. If you changed READY's colour with the `setcolor` serial command, it is your colour that blinks, between full and half strength. In BASIC, a colour so dark that READY's black icon would vanish at half strength, such as a deep red or blue, blinks to a lighter shade instead, and on a colour too dark for black writing, BASIC writes in white.

BASIC Mode Overview



BASIC Mode Click Commands

This is a new feature introduced in v2.5.4

Whereas the BASIC mode uses long clicks to validate the main commands you can use shorter click combinations for auxiliary actions in BASIC mode:



- You can **add a marker** with a unique ID to a station:

  LEFT-LEFT-LEFT-RIGHT

  > Markers will appear on the live map with their unique ID

- You can display the live **survey map**:

  LEFT-RIGHT-LEFT-RIGHT

- You can access the **LRUD** entry mode:

  RIGHT-RIGHT-RIGHT-LEFT

- You can select the *survey direction* as **IN** :

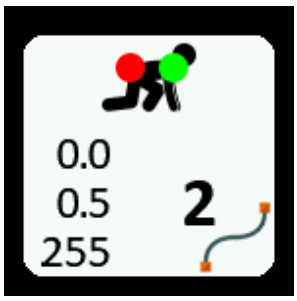
  LEFT-LEFT-LEFT-LEFT

- You can select the *survey direction* as **OUT** :

  RIGHT-RIGHT-RIGHT-RIGHT

> Changing the survey direction can only be done while in Stand By (Green) mode

> In Basic mode when you do short clicks you'll notice dots appearing at the top of the screen, RED for left clicks and GREEN for right, they are here to help you enter click commands



ECO Mode (Long Dives)

Introduced in v3.4.0.

ECO is a third survey mode, alongside BASIC and Verbose. Select it in:

OPTIONS > SETTINGS > SURVEY (BASIC *or* Verbose *or* ECO)

Each time you select that menu row the mode cycles **BASIC → Verbose → ECO → BASIC**. The row always shows the mode currently selected, and the choice is saved immediately.

It is built for one thing: **getting the longest possible survey out of one battery charge**. Everything you do is identical to BASIC — the same slider holds, the same click commands, the same wheel, the same readings. Only the screen changes, and the screen is where most of the power was going.

What it looks like

Instead of filling the whole display with the phase colour, ECO leaves the screen black and shows the survey phase as **one small coloured disc near the top of the screen**. The colours are exactly the ones you already know from BASIC:

Disc colour	Phase
Blinking **green**	STANDBY
White	READY
Red	STABILIZE
Flickering **blue / violet**	READING (the wheel is measuring)

(These are the default colours. If you changed them with the `setcolor` serial command, ECO uses your colours too.)

In READY the disc also shows whether the wheel is on the line *(v3.4.2+)*: slide the MNemo a little along the line, and while the wheel turns the disc blinks between white and a dimmer

white. See Is the wheel on the line?

The screen is also refreshed **twice a second instead of five times a second**. Together these bring the device down to **about 27 mA while surveying, against 55–57 mA in BASIC** — better than half the power for the same survey.

> **Your contrast setting is left alone in ECO.** An early version of the mode dimmed the panel as well, but ECO lights so few pixels that dimming them further saved very little — and it cost readability underwater, which is exactly where the mode is meant to be used.

> The click-code reminder is still shown in ECO. Dropping it would have saved about 1.5 mA, which is not worth losing the reference underwater.

The wheel is flashed, not lit *(v3.4.2+)*

The wheel is read by two small light sensors, which draw about **15 mA** when lit. BASIC and Verbose keep them lit for the whole of every shot. ECO now flashes them instead, **500 times a second**, each time just long enough for them to light up and be read: they are lit about an eighth of the time, and draw about **2 mA** while a shot is read. They also stay dark while the slider is read, so it reads as steadily as ever.

Nothing else changes. The blue/violet disc still flickers as the wheel turns, and the shot is counted exactly as with the sensors lit at any speed you can swim: a flash could only miss part of the wheel's turn above about 9 m/s.

ECO sections are named E##

A survey shot in BASIC is stored as `B01`, `B02`… A survey shot in ECO is stored as **`E01`, `E02`…**, so you can always tell afterwards — in the history list, in the web page and in the exported file — which surveys were shot in ECO, and therefore which ones may have ended in a flat battery. The two modes share one counter, so the numbers never collide.

ECO runs the battery all the way down

This is the part to understand before you use it.

In BASIC and Verbose the MNemo shuts itself down cleanly when the pack reaches the low-battery cutoff (7.0 V). That cutoff leaves a real amount of usable capacity unspent — sensible on a normal dive, wasteful on a long one.

In ECO the MNemo does not stop. The low-battery icon appears in the **top-right corner** of the screen — clear of the state disc and of the readings — as a *warning*, and the survey carries on until the pack is genuinely empty and the device dies.

That is the intended behaviour, not a fault. It is also why the next section matters.

If the power dies mid-survey, your shots survive

Because ECO is designed to end in a power loss, the firmware now expects it.

If the MNemo loses power in the middle of a section — flat battery, flooded compartment, anything — the section is left without its end marker. On the **next power-on** the device closes that section for you, before it scans the memory, and displays:

> **Survey recovered**

The shots you had already taken are all there: in the history, on the map, and in your download, as an ordinary finished section.

> **This used to lose data.** Before v3.4.0 a section cut short like that survived only as long as it was the last thing in memory. As soon as you started your next survey on top of it, **the whole dangling section was discarded**. That is fixed.

When to use which mode

	Use
BASIC	Normal dives. The recommended everyday mode — full-screen colour, easiest to read at a glance.
ECO	Long dives, or any dive where running out of battery is a real possibility.

	Use
Verbose	When you need the options only Verbose offers, and you can read the screen. *(legacy)*

You can switch modes at any time from the menu between surveys.

Verbose Mode (legacy)

Starting a survey

From the main menu select **SURVEY** The screen will display options similar to:



In this menu you can choose the name of the section you are surveying.

Section in this case means the section of the survey which can differ from the section of the cave you are working in.

In our example the section is called AA1. You can navigate from one letter to the other moving the slider to the *right* and change the current letter by moving the slider to the *left*. Once done select OK.

The menu will give you the choice of the direction you are surveying, IN or OUT.

- **IN** means you are going from already surveyed stations to new ones
- **OUT** means you are coming from new stations going towards already known ones.



> NOTE : The reference shot that you had in Mnemo v1 are no longer used in v2

Surveying a shot

To survey a shot:

- pull the pressure plate lever
- insert the line into the slot
- release the lever
- make sure the line is not trapped
- slide the device to the beginning of the shot (Entry Station)

Select **START**



The display will show a *RED* square:



At that point the device is waiting to be stabilized to measure the azimuth.

> **THE ONE THING TO REMEMBER FROM THIS USER MANUAL:** Don’t hold the device in your hand, rather support it from beneath so it can use the tension in the line to self align and take the most accurate measurement possible.

The display shows a *BLUE* or *PURPLE* square with a dot in it when the compass has measured the first azimuth.



You can now move the device along the line all the way to the next station. The display will be flickering that BLUE/PURPLE square indicating that the wheel is spinning and measuring the length of the line. The device measures the line in both direction allowing you to compensate for big diameter tie-offs. *If the line is too loose the length reading will be incorrect.*

> You can use Mnemo v2 in reverse direction of the engraved arrow, the device will automatically reverse the azimuth
>
> If the device is flickering BLUE/RED that means it has been dragged along the line before the stabilization phase was over, you should start the survey of that shot again.

When you reach the next station, select **STOP**

The display turns red waiting to stabilize and measure the second azimuth.

Repeat the operation …

Once you reached the last shot of your section select **END**

This will save the section. You are back to the main menu ready to survey the next section !

Cancelling / Retaking a measure

If you want to retake a measure, there are two possibilities:

- Either you are still in reading mode (BLUE/PURPLE background), in this case just select BACK and you can retake the measure you were taking.
- You finished the measure (WHITE background waiting for next measure), in this case you can select UNDO to erase the last saved measure and take it again.

Recording the surface pressure

The device calculates the depth by measuring the variation of pressure between the surface and where it is located under water. Therefore it needs to have a reference measurement of the surface pressure. This can be done by navigating to :

OPTIONS > SENSORS > DEPTH > SURFACE

Once you get the *CALIBRATED* message on the screen that means the device recorded the current atmospheric pressure as reference.

> This needs to be recalibrated mainly when diving at altitude or diving during extreme weather conditions

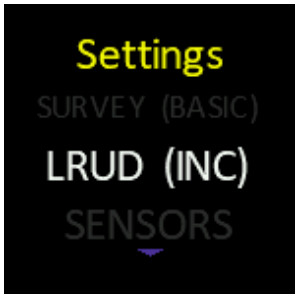
> Obviously ... ***DO NOT ADJUST SURFACE PRESSURE IN THE WATER***

LRUD (Cave Profile description)

You can enter LRUD (Left-Right-Up-Down) measurement (or estimation) values to describe the corridor section of the cave at each station.

- This is done by going to

  OPTIONS > SETTINGS > LRUD(LIST or INC)

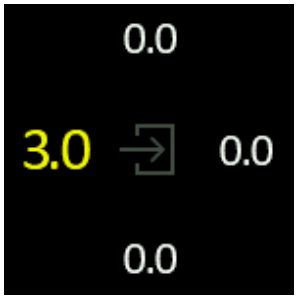
  

  - ***INC*** : In this mode the numerical value will increase gradually. It is the slowest of the two modes but allows to enter precise values
  - ***LIST*** : In this mode the numerical value will increase according to a predetermined set of lengths. It's the fastest mode and typically the one used when value are estimated rather than measured.

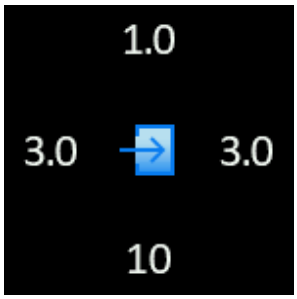
- In BASIC Mode, you can also enter the LRUD menu for chosen stations when you are on the READY screen (the one with WHITE background) with a click combination:

  RIGHT-RIGHT-RIGHT-LEFT

Once you are on the LRUD entry screen you increase the value moving the cursor left and go to the next value moving it right.



Finally you exit the screen by selecting the button in the middle :



Interference Marker

During your survey, if you see an icon of a magnet flashing on your screen, that means the device has detected a magnetic interference.



That flashing will only last for 5s, when it stops it doesn’t mean the problem is gone, just that you were warned. In such a case make sure that no part of your equipment has magnets or other elements that could strongly interfere with the compass.

About Stabilization

In which ever survey mode you use the Mnemo the **stabilization** phase is indicated by **RED background color**.

If during stabilization the red background starts flashing, that means the device is not flat enough (ie. rolling too much on one side or the other). In this case rotate the device around the line until the blinking stops, then stay stable and the measure will be taken.

The rotation tolerance is quite large (about 30º on each side of the line) so this should not be a major issue.

Surveying - The big picture -

An important thing to remember is that once you are using MNemo it is not necessary to read the display for each operation. The color will indicate the operational phase. Familiarity with MNemo allows your focus to be on the safety of your dive and a global cave/environment awareness. BASIC mode makes it even easier, we recommend that mode for all your surveys.

With a bit of practice you can survey at the same speed you’d normally swim in a cave (15-20m/min).

Survey History and Direct View

You can access the history of your surveys on the device in the Menu at:

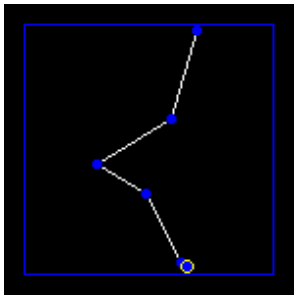
OPTIONS > HISTORY

It will display a list of all the surveys on your device. By selecting one of those surveys you'll get access to a map of that survey as well as some basic statistics.

> **Survey names.** Surveys shot in BASIC mode are named `B01`, `B02`... and, *since v3.4.0*, surveys shot in ECO mode are named `E01`, `E02`... so you can tell afterwards which ones were shot in ECO — and therefore which ones may have ended on a flat battery. Both modes share a single counter, so the numbers never repeat. A survey named by hand in Verbose mode keeps the name you gave it.



In BASIC and ECO mode, you can also bring up the latest survey — or the survey currently being done — at any time with the **LEFT-RIGHT-LEFT-RIGHT** click command (details here). It displays a map that turns with the device.



Give an impulse left to the slider button (SELECT) to go on to the 3D view, then to a screen with the survey's statistics, and once more to leave (see **2D / 3D Map View** below).



> The double tap on the back of the Mnemo that used to open this view has been removed; see the firmware changelog. In Verbose mode, you can see a section's map in the history once you have ended it.

2D / 3D Map View

When a map is displayed — whether from History or from the BASIC mode shortcut — pressing the **select button** cycles through three views:

```
2D Map  →  3D Map  →  Statistics  →  exit
         [select]    [select]       [select]
```

3D Map

The 3D view renders the survey in perspective with depth-based colour: **blue = deepest, red = highest** (since v3.1.0; the on-device gradient now matches the web view's). Three reference axes are shown at the origin:

- **Yellow** arrow — vertical (UP)
- **Red** arrow — North
- **Blue** arrow — East

The camera angle is controlled by the **IMU** — simply tilt and rotate the device to change the viewing direction. The camera framing adjusts automatically to fit the entire survey in view.

Web Interface

When connected over WiFi (see Wireless Data Transfer), every recorded survey can be viewed in your browser as well as on the OLED.

2D survey map

From the device's main web page, clicking a survey opens the `/View` page: a top-down SVG map of the survey path. *(v3.1.0+)* The path is drawn as per-segment lines coloured by depth — **blue = deepest, red = highest** — so it stands out clearly against the dark background. Stations are marked with circles (yellow at the start, red elsewhere).

Interactive 3D viewer *(v3.1.0+)*

On the 2D map page a ▶ **View in 3D** button opens `/View3D`, a full-screen interactive 3D viewer of the same survey. Controls:

Action	Mouse	Touch
Orbit	Left-drag	One-finger drag
Dolly (camera distance)	Right-drag up/down	—
Zoom	Scroll wheel	Pinch
Re-centre on a station	Click the station	Tap the station

A base-plane grid is drawn at the deepest survey point, and a small compass-rose axes widget (East / North / Up) sits in the lower-left corner. Use the **← 2D** link at the top-left to return to the 2D map.

Recovered surveys *(v3.4.0+)*

If the device lost power in the middle of a section, the section is closed at the next power-on — the device shows **Survey recovered** — and it then appears in this list like any other finished survey, with the shots that were already recorded. See Memory management.

Memory management

Mnemo can save around 5000 measures. You can see the amount of memory used on the bottom bar in the main menu.

You can erase the memory by going to: **OPTIONS > SETTINGS > SYSTEM > RESET MEMORY**

> We recommend downloading the data from the MNemo after each survey session, backing up the data as DMP and erasing the data from the device in order to start next session with an empty memory.

A DMP you have backed up can be turned into a spreadsheet at any time — drop it on converter.arianesline.com. See Converting a DMP to Excel.

Erasing from the web interface *(v3.4.0+)*

The memory can also be erased from the device's built-in web page, without going through the device menu. On the main page, the **Erase memory** button opens a confirmation page which first shows you what you are about to lose:

- the number of **surveys** and **shots** currently stored, and the **percentage of memory in use**;
- a **Download DMP** and a **Full CSV** link, so you can save the data before erasing it.

The erase only happens once you type `ERASE` in the confirmation box and submit the form. Reloading or bookmarking the page cannot wipe your data.

> The page refuses the erase while the device is busy — mid-survey, calibrating, or in depth-gauge, test or Mr Magneto mode. Finish or leave what the device is doing and try again.

Device settings and stored WiFi networks are not affected; only survey data, the marker counter and the section counters are cleared.

If the device loses power mid-survey *(v3.4.0+)*

A section interrupted by a power loss — a flat battery in ECO mode, a flood, anything — is closed automatically at the next power-on. The device shows **Survey recovered** and the shots already recorded appear in the history, on the map and in your download as an ordinary finished section.

> Before v3.4.0 such a section survived only while it was the last thing in memory; starting a new survey on top of it discarded it entirely.

Setting the water type

You have the choice between fresh and salt water and you can access the option by navigating in the menu to :

OPTIONS > SENSORS > DEPTH > WATER *(fresh or salt)*

Device Settings

Display and device preferences are grouped under:

OPTIONS > SETTINGS > DEVICE SETTINGS

Screen Orientation

Navigate to **OPTIONS > SETTINGS > DEVICE SETTINGS > LCD ORIENT.(X)**

Each time you click on the menu entry the screen rotates 90 degrees.

Display Contrast

Navigate to **OPTIONS > SETTINGS > DEVICE SETTINGS > CONTRAST**

The contrast value goes from 2 to 16, with 16 being the brightest.

Language

Navigate to **OPTIONS > SETTINGS > DEVICE SETTINGS > LANGUAGE**

The on-device interface is available in six languages:

- **ENGLISH**
- **ESPANOL**
- **FRANCAIS**
- **DEUTSCH**
- **ZH (SIMP)** — Simplified Chinese (Mainland)
- **ZH (TRAD)** — Traditional Chinese (Taiwan)

The currently active language is marked with a `<-` arrow next to it. Select a different entry to switch languages immediately. The setting is saved to device memory and persists across power cycles.

Language

The MNemo on-device interface is available in six languages. Navigate to:

OPTIONS > SETTINGS > DEVICE SETTINGS > LANGUAGE

You will see the list of available languages:

- **ENGLISH**
- **ESPANOL**
- **FRANCAIS**
- **DEUTSCH**
- □□□□ — Simplified Chinese (Mainland)
- □□□□ — Traditional Chinese (Taiwan)

The currently active language is marked with a `<-` arrow. Select a different entry to switch immediately — all on-device menus, prompts and feedback messages update at once.

The language setting is saved to device memory and persists across power cycles. It is independent of the unit system or any other setting.

Date and Time Synchronization

The MNemo v2 stores the date and time of each survey. It is important to ensure the device's clock is correctly set before going on a dive, as an incorrect date/time will affect the timestamps recorded in your survey files.

How to Synchronize the Date and Time

From the web interface (recommended)

The easiest way to set the clock is from the device's built-in web page over WiFi — no extra software beyond a browser once WiFi is available.

1. Turn WiFi on (see Wireless Data Transfer) and open the Mnemo's page in your browser. *(v3.1.0+: scan the QR code shown on the OLED.)*
2. Open the **Settings** page from the top menu.
3. Under **Time & Format → Date & time**, tap **Synchronize date & time**.

The device clock is set to the **current date and time of the browser** (i.e. your phone or computer). Two taps and you're done.

(v3.4.0+) The page now **tells you the result**: it confirms that the device clock was synchronized, or says plainly that the write failed or that the request was rejected. Before v3.4.0 a sync that did not take looked exactly like one that succeeded — if it reports a failure, try again, and check the clock on the Main Menu status line before diving.

> **Tip:** Make it a habit to open the Mnemo page and tap the sync button at the start of every dive trip.

Over the USB serial console (advanced)

Power users can also sync the clock over USB by sending the `syncdatetime` command on the device's serial port (57600 baud) followed by 5 bytes (year minus 2000, month, day, hour, minute). This is the protocol that desktop helper tools use under the hood.

Checking the Current Date and Time

You can verify the current date and time directly on the device from the **Main Menu**. The bottom status line shows:

Display	Description
26/03 11:26 27.6ºC	*Day/Month Hour:Minute Water Temperature*

The Day/Month order can be swapped to Month/Day on the web Settings page (**Time & Format → Date format**).

If the date and time look incorrect, follow the steps above to resynchronize.

Potential Issues When the Date Has Not Been Synchronized

If the device's clock has not been synchronized, the following issues can arise:

- **Incorrect survey timestamps**: each survey is recorded with a date and time. If the clock is wrong, it will be difficult to match a survey to the date of the actual dive.
- **Default date fallback**: if the internal clock has lost its settings (for example after a fully depleted battery), the device falls back to a default date. A **DATE RESET** warning is shown for a few seconds when the Mnemo is turned on to alert you.
- **Data-management confusion**: when reviewing surveys in your cave-survey software, incorrect dates make it harder to organise and identify dives, especially when multiple surveys are stored on the device.

> **Important:** if you see the **DATE RESET** warning when turning on your MNemo, resynchronize the clock from the web interface before your next dive.

Battery

- The Mnemo contains a 2S - 450 mAh sealed lipo battery.
- The Mnemo can be recharged using either the USB port of your computer or a dedicated charger. It requires no more than 150mA charging current.
- The red LED indicates that the charge is in progress. It will turn off when the charge is complete.





It is recommended to have the Mnemo turned OFF before connecting for a charge.

> Note that if the device is ON and connected to a charger, it will not turn off when you select EXIT in the main menu but simply go into Energy Saver mode (Blinking Battery on the screen)

Making the charge last *(v3.4.0+)*

Three changes in v3.4.0 stretch a single charge, two of them without you doing anything:

- **The processor rests between jobs.** The MNemo used to run at full clock for an entire dive even when there was nothing to do. It now halts between scheduled jobs and wakes on the next one — or immediately on a wheel tick or a button press. Nothing looks different; the battery simply lasts longer.
- **WIFI OFF really turns the radio off.** Switching WiFi off used to leave the radio powered and still associated, holding the device at roughly **80 mA instead of its 28 mA idle**. The radio is now genuinely powered down.
- **ECO survey mode** more than halves the current drawn while you survey — about **27 mA against BASIC's 55–57 mA** — by replacing the full-screen colour with a single small coloured dot on a black screen.

From **v3.4.2**, ECO also stops lighting the wheel through every shot: it flashes the wheel's sensors instead, so they draw about 2 mA rather than about 15 mA while you measure. See The wheel is flashed, not lit.

Low battery

When the pack runs low, a **low-battery icon** appears on the screen and the display is dimmed.

- In **BASIC** and **Verbose** the MNemo then shuts itself down cleanly at the cutoff, ending the current section first.
- In **ECO** it does **not** — the icon is a warning, and the survey deliberately continues until the pack is empty. If the device dies mid-section, the next power-on reports **Survey recovered** and your shots are kept. See ECO Mode.

Tools

Navigate to: **OPTIONS > TOOLS**

You'll find a list of diagnostic and analysis tools:

- **Depth Gauge**: A depth gauge indicating depth and absolute pressure
- **Test Mode**: A mode to test the reading of all sensors — pressure, magnetometer, inclinometer, etc.
- **ONE-GO(OFF)**: *NOT DOCUMENTED, leave on OFF*
- **MAGNETO**: Magnetic analysis tool

> WiFi Access Point mode has moved to **OPTIONS > WIFI > WIFI AP**. See Wireless Data Transfer for details.

Magneto

You can use the Magneto feature to analyse how strongly your dive equipment affects magnetic readings.

> The earth magnetic field ranges from 25 µT to 65 µT depending on your location.

By moving the Mnemo closer to or further from your equipment you'll get a reading of the strength of the magnetic field created by that equipment.

- Here you can see that the magnetic field is between 32.92 and 33.44 when the device is left without any outside disturbance



- When we move a dive light around the Mnemo the magnetic field is this time fluctuating between 30.55 and 37.49 — a huge difference!


Magneto
>RESET< EXIT
Bc= -17.58,-17.50,-26.31
Gp= -0.02,0.00,-1.01
delGp= 0,1,-1
30.55<uT= 36.16<37.43
avg: 33.96 #:165
V2
MN

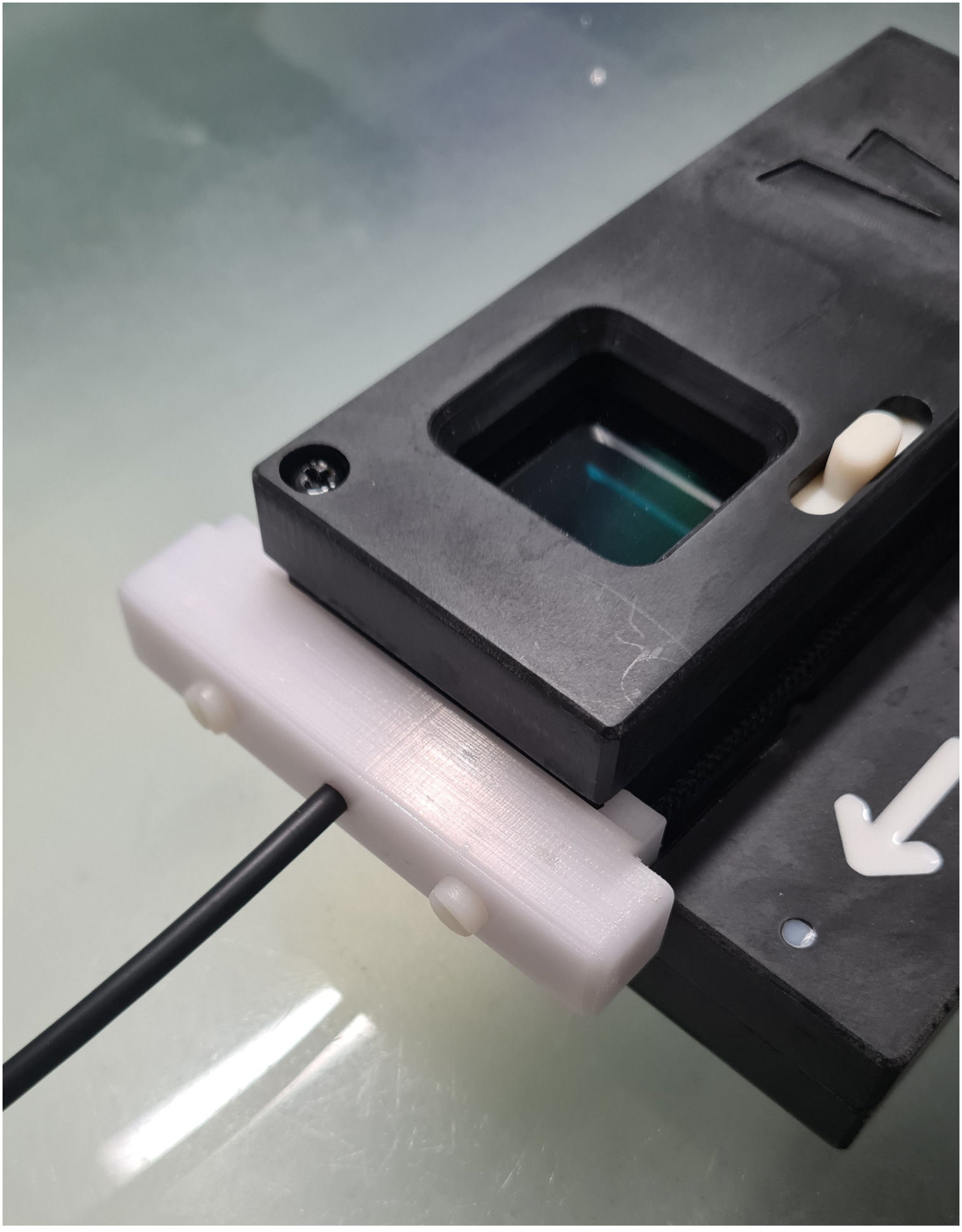
USB Connection

The Mnemo v2 is supported on Microsoft Windows(>7), Apple Mac OSX (>10.6) and Linux (Kernel >5.0.0)

Currently there’s no driver installation required for the Mnemo v2 on any of the supported operating system.

Connection

Connect the USB cable to the device as shown on the picture and then to your computer. The order is important.



The blue led indicates that the connection to the computer/charger was successful.





Turn on MNemo. At that point your computer should have recognized the device.

Downloading your surveys

Ariane can read the survey data directly from the device over USB. Use the latest version of Ariane.

> Downloading over WiFi, from the device's web page, is the recommended method: see Wireless Data Transfer.

The DMP data from the Mnemo can be converted online at converter.arianesline.com, to an Excel workbook, CSV or JSON, with nothing to install: see Converting a DMP to Excel.

What else you can do over USB

- Update the firmware: see Firmware Update.
- Read data, change settings and run diagnostics from a terminal: see Serial CLI.

Serial Command Line Interface

The Mnemo exposes a **serial command-line interface (CLI)** over its USB connection. It lets you read data, change settings, and run diagnostics from a computer — including several settings that are not available from the on-device menu.

Most users will never need this: the on-device menus and the device's web page cover everyday use. The CLI is documented here for advanced users and troubleshooting.

Connecting

1. Connect the Mnemo to your computer with the USB cable (see USB Connection).
2. Open any serial terminal program and connect to the Mnemo's serial port at **57600 baud**.
3. Type a command name, optionally followed by arguments, and press **Enter**.

Commands that produce a serial response finish it with an end-of-transmission marker, so host software knows when the response is complete. A few commands that only change a setting or update the display return without emitting a marker.

> **TIP**
>
> Not sure what's available? Type `help` to list every command with its description, or `help <command>` for a single one. `listcommands` prints just the command names.

> **DANGER**
>
> Commands marked **[DO NOT USE]** are low-level factory, hardware, and debug operations. They can corrupt device settings, survey data, or the hardware itself. Leave them alone during normal use.

Data transfer

Command	Arguments	Description
`getdata`	`[address(>800)]`	Reads all survey data from the device, or reads until the optional memory address is reached.
`syncdatetime`	—	Synchronizes the device date/time with the computer's.

Help & discovery

Command	Arguments	Description
`help`	`[command]`	Shows help for all commands, or for a single named command.
`listcommands`	—	Lists the names of all available commands.

Device identity & firmware

Command	Arguments	Description
`setname`	`name`	Sets the custom (friendly) name of the device: 1 to 19 letters (A–Z, a–z), digits, spaces, hyphens or underscores. Spaces at either end are dropped. Any other name is refused with an error line, and the current name is kept.
`getname`	—	Gets the custom name of the device.
`getfirmwareversion`	—	Gets the firmware version installed on the Mnemo.

Command	Arguments	Description
`getwififirmwareversion`	—	Gets the firmware version installed on the WiFi (NINA) module.

WiFi network management

Command	Arguments	Description
`addwifinet`	`NetworkName Password`	Adds a WiFi network to the device, or replaces the password of one already saved under that name: a name of 1 to 32 bytes, and a password of 8 to 63 characters (an accented letter counts as 2), or none for an open network. If either contains a space, put both in double quotes: `addwifinet "My network" "my password"`. Up to 20 networks are kept. A network that does not fit is refused with an error line saying why, and nothing is saved. An error line also reports a network the WiFi module's storage did not keep.
`removewifinet`	`NetworkName Password`	Removes a WiFi network from the device.
`clearwifinet`	—	Clears the entire list of saved networks.
`listwifinet`	—	Reads the list of saved networks.
`checkwifistorage`	—	*(v3.4.1+)* Checks that the device can still save networks. Prints `WiFi storage: OK` or `WiFi storage: FAILING`.
`repairwifistorage`	`[force]`	*(v3.4.1+)* Runs the same check and, only if it fails, clears the WiFi radio's storage area so that networks can be saved again. **Deletes the saved networks.** `force` clears it even when the check

Command	Arguments	Description
		passes. Prints the same result as the menu item, such as `WiFi storage: Repaired`. See Saved networks are not kept.

Click / button tuning

Command	Arguments	Description
`getclickthreshold`	—	Gets the click threshold percentage.
`setclickthreshold`	`level`	Sets the click threshold percentage, 10–90; any other value is ignored.
`getunclickdelta`	—	Gets the unclick threshold percentage.
`setunclickdelta`	`level`	Sets the unclick threshold percentage, 2–40; any other value is ignored.
`getBMclickfactor`	—	Gets the Basic Mode click time factor (in %).
`setBMclickfactor`	`factor`	Sets the Basic Mode click time factor (in %).
`activatesafeon`	—	Activates the safety-to-turn-on feature.
`deactivatesafeon`	—	Deactivates the safety-to-turn-on feature.

Compass & stabilization

Command	Arguments	Description
`getstabilizationfactor`	—	Gets the stabilization factor.

Command	Arguments	Description
`setstabilizationfactor`	`factor`	Sets the stabilization factor.
`getmagneticinterferencethreshold`	—	Gets the tolerated magnetic-interference threshold (in %).
`setmagneticinterferencethreshold`	`value`	Sets the tolerated magnetic-interference threshold (in %).

Display colors

Command	Arguments	Description
`getcolor`	`state`	Gets the color of a state (`standby`, `readinga`, `readingb`, `stabilize`, `ready`).
`setcolor`	`state color24b`	Sets the color of a state to a 24-bit hex color (e.g. `setcolor ready 354250`).
`defaultcolorscheme`	—	Restores the default color scheme.

Settings & reset

Command	Arguments	Description
`defaultsettings`	—	Restores all default settings **and erases the memory**.
`testmode`	—	Puts the device into Test mode.

Runtime statistics

Command	Arguments	Description
`gettimeon`	—	Gets the time (in minutes) the device has been switched ON.
`gettimesurvey`	—	Gets the time (in minutes) the device has been surveying.

Over-the-air update

Available on firmware built with OTA support.

Command	Arguments	Description
`otaupdate`	`url`	Downloads an application `.bin` from an `http://`/`https://` URL and self-applies it (reboots; WiFi must be connected).
`otaupdatenina`	`url`	Flashes the NINA radio firmware from a `nina.bin` URL (releases WiFi, then flashes — takes several minutes).

See Firmware Upgrade for the recommended update procedure.

Low-level / factory — [DO NOT USE]

Command	Arguments	Description
`picbootmode`	—	Prepares the device for PIC reprogramming.
`getpicinfo`	—	Gets PIC hardware/firmware information.
`picprog`	`[INIT]`	Uploads the in-memory firmware onto the PIC. Pass `INIT` for the initialization variant.

Command	Arguments	Description
`getpicswitchonthreshold`	—	Gets the PIC switch-on threshold.
`setpicswitchonthreshold`	`value`	Sets the PIC switch-on threshold to the given numeric value.
`eepromwrite`	`address byte`	Writes a byte to a specific EEPROM address.
`eepromread`	`address`	Reads a byte from a specific EEPROM address.
`readfile`	`filename`	Reads a file stored on Flash memory.
`erasefile`	`filename`	Erases a file stored on Flash memory.

UI-test builds add extra commands (`screencap`, `inject`, `gotomain`) used only for automated interface testing; they are not present in production firmware.

Wireless Data Transfer

WiFi settings are now grouped in their own submenu at **OPTIONS > WIFI**, which contains:

- **WIFI ON/OFF** – connect to a known network
- **WIFI AP** – start a local Access Point hosted by the device
- **WIFI ON AT START** – auto-connect on every power-on
- **Back**

Each row shows the state it will put the device in. *(v3.4.0+)* While you are connected to a local network the AP row reads **WIFI_AP (SWITCH)** rather than `(OFF)` — selecting it there does not add an access point alongside the network, it drops the network connection and switches the device over to its own AP.

> **WIFI OFF really powers the radio down** *(v3.4.0+)*. Until v3.3.1 switching WiFi off left the radio powered and still associated, quietly holding the device at about **80 mA instead of its 28 mA idle**. Turning WiFi off now takes the radio out of the power budget; the next WiFi command brings it back up on its own. See Battery.

Local Network

Adding networks

Before the Mnemo can connect to your local network, it needs that network's name and password. You give them to it through the Mnemo's own WiFi access point:

1. On the Mnemo, go to **OPTIONS > WIFI** and select **WIFI AP**. The Mnemo starts its own WiFi network, named after the device (**MNemo** unless you renamed it), with the password **password**.
2. On your phone or computer, join that network. Then open a browser at the IP address shown on the Mnemo's screen (normally `192.168.4.1`). *(v3.1.0+)* The Mnemo also shows QR codes that do both for you: see WiFi QR onboarding.
3. On the Mnemo's web page, press **Settings**. In the **Known networks** card, enter your local network's name under **SSID** and its password under **Password**, then press **Add network**. The page confirms with **Network saved.**

4. On the Mnemo, select **OPTIONS > WIFI > WIFI ON/OFF**, which reads **WIFI (AP)** while the access point is on. The Mnemo closes its access point and connects to your local network: see Connecting.

> The local network must operate on **2.4 GHz**: the Mnemo cannot connect to a 5 GHz network. If your router gives its 2.4 GHz and 5 GHz networks different names, add the 2.4 GHz one.

The Mnemo keeps up to 20 networks. The same **Known networks** card removes a network, or all of them. Over USB, the `addwifinet` serial command adds a network too.

If networks you add are gone the next time you turn WiFi on, see Saved networks are not kept.

Connecting

On your Mnemo navigate to **OPTIONS > WIFI > WIFI ON/OFF**:



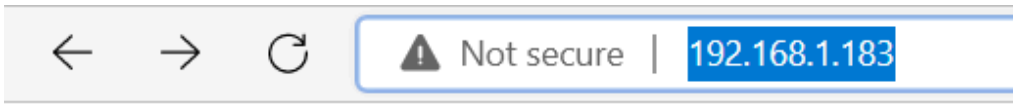
The Mnemo will try to connect to its known networks. If successful it returns to the main menu and **displays the IP address of the device**.



> The icon in the top left corner indicates the strength of the wireless connection between the router and the MNemo.

Open a browser and navigate to the displayed IP address to preview surveys or download your data. This is the recommended way to download your surveys. Ariane can also read them directly from the device over USB: see USB Connection.

> The DMP you download here can be converted to Excel at converter.arianesline.com — no software to install.



MNemo

SN: 009315501C759D60

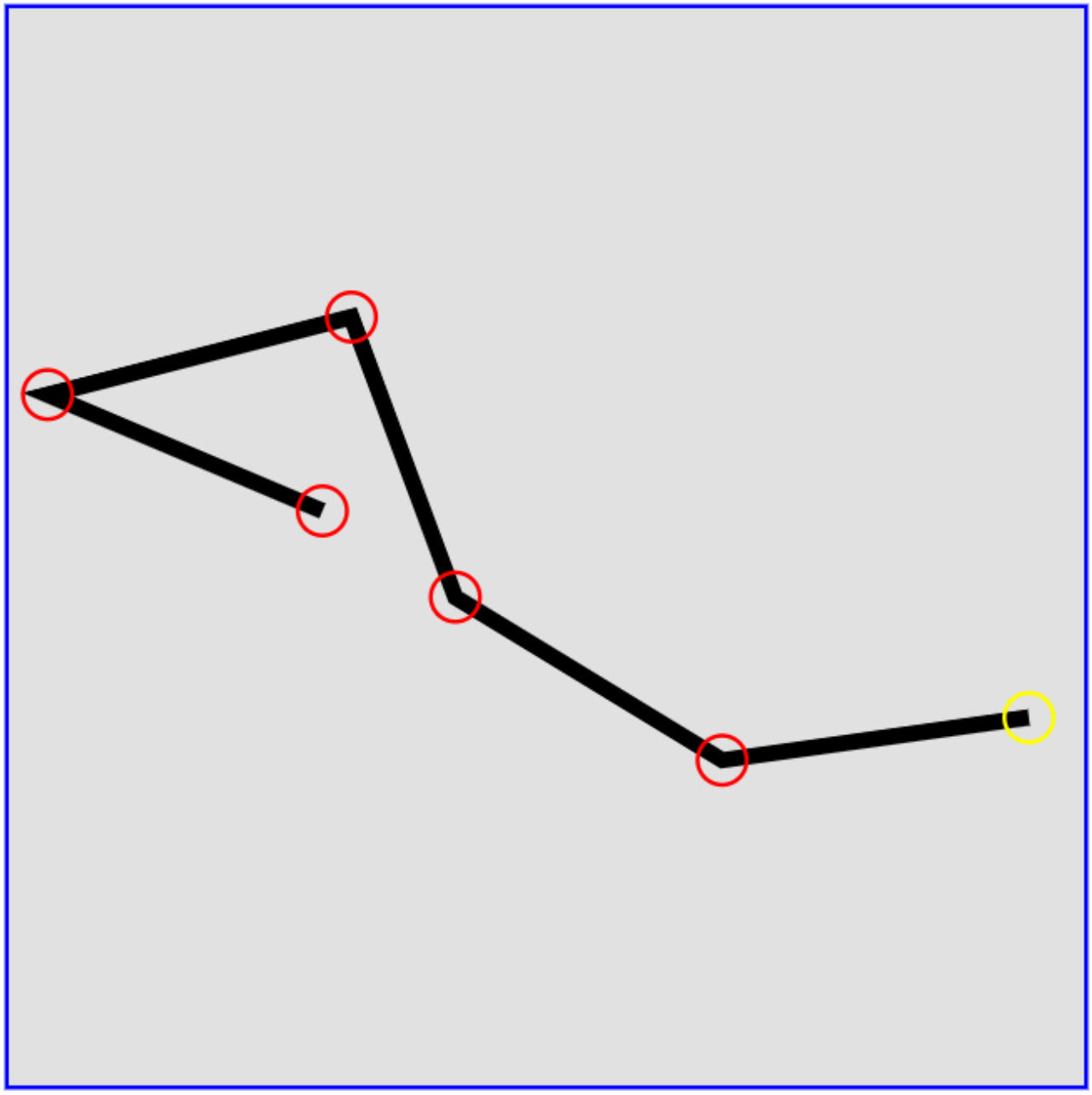
Download DMP
0-BAS-0/8-#2
1-BAS-0/8-#5

Not secure | 192.168.1.183/View?id=1

MNemo

SN: 009315501C759D60

1-BAS-0/8-#5



Converting a DMP to Excel

A DMP file downloaded from the device's web page can be turned into a spreadsheet without installing anything:

converter.arianesline.com

Drop the `.dmp` on the page and you get back an Excel workbook with a summary of the survey, one sheet containing every shot, and one sheet per section. CSV and JSON are available too. The file is converted on the spot and is never stored.

WiFi at Start

Navigate to **OPTIONS > WIFI > WIFI ON AT START** and toggle the option on.

When enabled, the device automatically starts its **WiFi Access Point** every time it powers on. The Access Point shuts itself off after **30 seconds** if no one connects to it, so it doesn't drain the battery when unused.

> Combined with the QR onboarding below, this makes opening the Mnemo's web page on your phone right after switching the device on a two-scan operation.

Not underwater *(v3.4.0+)*

WiFi does not travel through water, so an access point started underwater can only drain the battery. From v3.4.0 the device therefore:

- **skips** the automatic access point when it is switched on while submerged, and
- **shuts it down** within about 5 seconds if the device goes underwater while the automatic access point is up.

> This applies **only** to the access point started automatically by *WiFi At Start*. WiFi that you switched on yourself from the menu is never turned off for you — dive or no dive.

WiFi QR onboarding *(v3.1.0+)*

When you turn WiFi on, the Mnemo shows a **QR code** on the OLED so you don't have to type the network name, password or IP address on your phone.

Local network (WIFI ON/OFF)

Once the Mnemo connects to your network it shows a QR code containing the page address (`http://<device-ip>`). Open your phone or computer camera app, point it at the screen, and tap the prompt to open the Mnemo's web page directly.

Access Point (WIFI AP)

In AP mode the onboarding happens in two stages:

1. **Join QR** — encodes the AP credentials (the device's name as SSID, `MNemo` unless you renamed it, and `Password: password`). Point your phone's camera at it and tap the prompt to join the Mnemo's network — no typing.
2. **Open-page QR** — appears automatically once your phone associates with the AP. Scan it to open the Mnemo's page in the browser. The QR dismisses itself the moment the page actually loads, returning the OLED to its normal screen.

You can dismiss either QR at any time with the device's **NEXT** or **SELECT** buttons.

The Mnemo as Wireless Access Point

If your local network is not accessible you can configure the MNemo as a Wireless Access Point. It is also how you give the Mnemo your local network in the first place: see Adding networks.

Navigate to **OPTIONS > WIFI > WIFI AP** and select the entry.

You can then connect your computer (or phone) to the wireless network created *(SSID: the device's name, **MNemo** unless you renamed it; Password: **password**)* and navigate to the displayed IP address as you would on a local network.

Erasing the memory from the web page *(v3.4.0+)*

All recorded surveys can be erased from the web interface, after a confirmation page that shows what is stored and offers a download first. See Memory management.

Saved networks are not kept *(v3.4.1+)*

The Mnemo keeps its list of WiFi networks in a small storage area on its WiFi radio. On some older devices that storage area stops accepting changes: you add a network, but it is gone the next time WiFi turns on, and the list always comes back empty. Updating the firmware does not fix it, because an update never touches that storage area.

The device's web page tells you when this happens: after you add or remove a network there, a notice says that the change did not reach the WiFi radio's storage and will be lost at the next restart.

To check it and fix it, go to **OPTIONS > SETTINGS > SYSTEM > WIFI STORAGE** and select **CHECK & REPAIR**:

- The Mnemo first checks that it can still save, which takes about a second. If it can, the screen shows **Storage OK** and nothing is changed.
- If it can't, the screen shows **Not saving / Erasing store...**. The Mnemo clears the storage area, the radio sets it up again, and the Mnemo checks once more. This takes about 10 seconds. The result is **Repaired**, followed by **Add networks again**.
- If WiFi was on, it is turned off before the storage area is cleared.

After a repair the list of networks is empty. Add your networks again as described in Adding networks.

> The repair clears only the radio's own storage area. Your surveys, your settings and your compass calibration are not affected.

If the result is **Radio not supported**, the WiFi radio is running firmware older than 1.4. Update it first: see Firmware Update. **Radio not answering** means the radio did not respond at all: switch the Mnemo off and on, then try again. **Erase failed** or **Still failing** means the repair did not work. Report that, or a radio that keeps not answering: see Report an Issue.

The same check and repair are available over USB from the Serial CLI: `checkwifistorage` and `repairwifistorage`.

DMP File Format

Format version 5 — specification for third-party software

This page describes the content of `.dmp` files produced by the MNemo cave surveying instrument, so that survey applications can import them. It is self-contained: no access to the device or its firmware is required.

A DMP file is the MNemo's lossless export format. It contains every survey *section* recorded since the device memory was last erased, each section holding a sequence of fixed-size *shot* records. Files are obtained from the device's WiFi download page, and are named `MNEMOYYMMDDHHMMSS.dmp` (two digits per date/time component, date of the download).

The DMP data from the MNemo can be converted online at converter.arianesline.com, to an Excel workbook, CSV or JSON, with nothing to install.

---

1. File encoding

A DMP file is plain ASCII text: a single line of decimal integers separated by semicolons `;`, with a trailing `;` after the last value and no line breaks.

Each value represents **one byte** of an underlying binary stream, rendered as a **signed** 8-bit integer (−128…127). To recover the unsigned byte:

```
byte = value < 0 ? value + 256 : value        // i.e. value & 0xFF
```

Multi-byte fields are 16-bit **big-endian** integers spanning two consecutive byte values:

```
u16 = (byte[0] << 8) | byte[1]
s16 = u16 reinterpreted as two's-complement signed
```

Example: `4;-46;` → bytes `4, 210` → `(4 << 8) | 210` = **1234**. `-1;-42;` → bytes `255, 214` → 65494 → **−42** as a signed 16-bit value.

Parsers should tolerate surrounding whitespace and a missing trailing semicolon. All structure descriptions below refer to the decoded byte stream; offsets are in bytes.

2. Overall structure

```
DMP     := Section*
Section := SectionHeader ShotRecord* EOCRecord
```

There is no file-level header or trailer. Sections appear in recording order. A section is normally closed by an end-of-cave (EOC) record; a section interrupted by power loss may lack it (see §7).

3. Section header (13 bytes)

Offset	Size	Field	Value / meaning
0	1	Format version	`5`
1	3	Signature	`68, 89, 101`
4	1	Year	Two-digit year; add 2000 (e.g. `26` = 2026)
5	1	Month	1–12
6	1	Day	1–31
7	1	Hour	0–23 (time the section was started)
8	1	Minute	0–59
9	3	Section name	3 ASCII characters from `A`–`Z`, `0`–`9`
12	1	Direction	`0` = IN survey, `1` = OUT survey

The 4-byte sequence `5, 68, 89, 101` is the **section signature** — the only synchronization point in the stream. Parsers locate sections by scanning for it (see §7).

Direction records which way the surveyor was traveling: IN = entering the cave, OUT = exiting. It also determines which station the LRUD values describe (§5).

4. Shot record (35 bytes)

One record per survey shot (one leg between two stations). All 16-bit fields are big-endian; signedness as listed.

Offset	Size	Field	Type	Unit / meaning
0	3	Start code	—	`57, 67, 77`
3	1	Shot type	u8	`0` = CSA, `1` = CSB, `2` = STD, `3` = EOC (see §6)
4	2	headingIn	u16	Compass heading at the start of the shot, 1/10 degree (0–3599)
6	2	headingOut	u16	Compass heading at the end of the shot, 1/10 degree (0–3599)
8	2	length	u16	Distance traveled along the line, cm
10	2	depthIn	s16	Depth at the start of the shot, cm
12	2	depthOut	s16	Depth at the end of the shot, cm
14	2	pitchIn	s16	Vertical angle at the start of the shot, 1/10 degree (−1799…+1799)
16	2	pitchOut	s16	Vertical angle at the end of the shot, 1/10 degree
18	2	left	u16	Passage dimension to the left, cm (see §5)
20	2	right	u16	Passage dimension to the right, cm
22	2	up	u16	Passage dimension upward, cm

Offset	Size	Field	Type	Unit / meaning
24	2	down	u16	Passage dimension downward, cm
26	2	temperature	s16	Water temperature, 1/10 °C
28	1	hour	u8	Time of day the shot was completed
29	1	minute	u8	
30	1	second	u8	
31	1	marker	u8	`0` = no marker; `N` = user-set station marker number (see §5)
32	3	End code	—	`95, 25, 35`

Angles are stored in tenths of a degree (`1234` = 123.4°), distances and depths in centimeters, temperature in tenths of a degree Celsius.

5. LRUD and marker semantics

LRUD (left/right/up/down) values, when present, describe the passage cross-section at the station where the surveyor entered them:

- On **IN** sections, they belong to the station the shot *arrives* at (the shot's "out" end).
- On **OUT** sections, they belong to the station the shot *departs* from (the shot's "in" end).

Zero in all four fields means no LRUD was entered for that shot.

Marker numbers are assigned from a device-wide counter when the user tags a station during the survey. They generally increase in recording order, but the counter can be reset by the user, so numbers may repeat across sections. `0` means the shot carries no marker.

6. Shot types and the EOC record

Value	Name	Meaning
0	`CSA`	Legacy type from older devices; not produced by current devices
1	`CSB`	Legacy type from older devices; not produced by current devices
2	`STD`	Standard measured shot — the only data-carrying type in current files
3	`EOC`	End-of-cave: terminates the section

Importers must accept all four values, and should treat any type byte greater than 3 as `EOC`.

A section is closed by a full 35-byte record of type `EOC` whose data fields are **all zero**:

```
57;67;77;3;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;0;95;25;35;
```

The EOC record carries no measurement data and should not be imported as a shot.

7. Parsing rules

1. **Locate a section**: scan the byte stream for the signature `5, 68, 89, 101`. Bytes before a signature are garbage (e.g. remnants of an interrupted recording) and must be skipped.
2. **Read the header**: consume the 9 remaining header bytes (date, name, direction) immediately after the signature.
3. **Read records**: each must begin with `57, 67, 77` and end with `95, 25, 35`. On any mismatch, abandon the current section and resume signature scanning at the byte after the last valid position.
4. **End of section**: a record of type `3` (EOC) closes the section. A section may also end implicitly at end-of-file (unclosed section) or at a validation failure; shots already read remain valid.
5. **Empty sections** (a header followed immediately by an EOC record) may occur and can be ignored.

6. The start/end codes make accidental matches unlikely, but the format has no escaping: parse structurally (fixed 35-byte records) once inside a section, and only fall back to signature scanning after a validation failure.

8. Worked example

Two sections recorded the same day: an IN section `AB1` with two shots, then an OUT section `AB2` with one shot. A machine-readable copy of this example is available for download — dmp_file_format_sample.dmp — a useful first test case for an importer. Annotated (line breaks and comments added for readability — a real file is a single unbroken line):

```
5;68;89;101;          section 1 signature (version 5)
26;8;25;14;30;        2026-08-25 14:30
65;66;49;             name "AB1" (ASCII)
0;                    direction IN

57;67;77;             shot 1 start code
2;                    type STD
4;-46;                headingIn  = 1234 → 123.4°
4;-30;                headingOut = 1250 → 125.0°
2;20;                 length     =  532 → 5.32 m
4;-35;                depthIn    = 1245 → 12.45 m
5;30;                 depthOut   = 1310 → 13.10 m
-1;-42;               pitchIn    =  -42 → -4.2°
-1;-39;               pitchOut   =  -39 → -3.9°
0;120;                left  = 120 → 1.20 m
0;-6;                 right = 250 → 2.50 m
0;80;                 up    =  80 → 0.80 m
0;-106;               down  = 150 → 1.50 m
0;-111;               temperature = 145 → 14.5 °C
14;32;5;              shot time 14:32:05
0;                    no marker
95;25;35;             shot end code

57;67;77;             shot 2 start code
2;                    type STD
8;52;                 headingIn  = 2100 → 210.0°
8;37;                 headingOut = 2085 → 208.5°
2;-58;                length     =  710 → 7.10 m
5;30;                 depthIn    = 1310 → 13.10 m (continues shot 1)
5;-76;                depthOut   = 1460 → 14.60 m
-1;-60;               pitchIn    =  -60 → -6.0°
```

```
-1;-55;               pitchOut   =  -55 → -5.5°
0;90;                 left  =  90 → 0.90 m
0;-116;               right = 140 → 1.40 m
0;60;                 up    =  60 → 0.60 m
0;-36;                down  = 220 → 2.20 m
0;-112;               temperature = 144 → 14.4 °C
14;35;41;             shot time 14:35:41
1;                    marker #1 set at this station
95;25;35;             shot end code

57;67;77;             shot start code
3;                    type EOC (closes section 1)
0;0;0;
```

9. Older format versions

This specification covers format version 5, the version produced by current MNemo devices. Files from older firmware used lower version numbers with different, smaller record layouts and are not covered here. A version-5 parser identifies its sections by the exact signature `5, 68, 89, 101` and should report — not misparse — files whose sections carry a different version byte.

Firmware Update

The MNemo can update its firmware **over WiFi (over-the-air)** or **over USB**. Over-the-air is the easiest once a device has been set up for it; the USB method below always works and is needed for the one-time switch-over.

Update over WiFi (over-the-air)

From **v3.2.0**, the MNemo can download and install new firmware by itself, over WiFi — no cable. It updates both the **main firmware** and the **WiFi radio**, and it works **on battery**.

- Turn WiFi on. If a newer version is available, an **update available** prompt appears — accept it. You can also start an update at any time from **OPTIONS > SETTINGS > SYSTEM > OTA UPDATE**.

  > *(v3.4.0+)* The check runs on the **first WiFi connection after each power-on**, so switching WiFi off and on again in the same session does not re-check. If you dismissed the prompt and want it back, either power-cycle the device or use **OTA UPDATE** from the menu.

- The MNemo connects to a saved WiFi network, downloads the update, and asks you to **confirm**.
- When prompted, **hold SELECT** to restart and apply. On battery, keeping SELECT held keeps the device powered across the restart.
- After it restarts, the **What's New** screen shows what changed.

> **One-time USB setup.** A device that has never used over-the-air updates (anything before **v3.2.0**), or one still on the older 1 MB update area, must first be flashed **once over USB** with a current version (see *Manual update* below). That installs the updated bootloader. After that single USB update, every future update — including the larger builds with Chinese fonts — arrives over WiFi.

What's New screen

After every update — over WiFi or over USB — the MNemo shows a short **What's New** page on the first boot, listing the release's headline features. Press **SELECT** to page through it. You only

see it **once per update**, and you can read it again any time from **OPTIONS > SETTINGS > SYSTEM > WHAT'S NEW**.

Manual update

- Download the latest firmware on Github (It’s a file with a .UF2 extension)
- Connect the device to your computer (see USB Connection) and go to:

OPTIONS > SETTINGS > SYSTEM > UPDATE

> The device should appear in your file explorer as a USB Memory stick would.

- Simply copy the firmware file you downloaded there. That should trigger a reboot of the Mnemo and install the new firmware.
- After updating the firmware, disconnect from the computer and turn off the MNemo.

> The next time you turn the Mnemo on the new firmware will be fully functional.

Maintenance

- Do not let the battery drain completely. Put it on charge (leave connected to your computer or other USB charger) once per month even if you don’t use the device.
- After each use thoroughly rinse MNemo with fresh water, do not use any other chemical product. Ideally have MNemo clipped on a piece of line and move it forward and backward so that the wheel gets cleaned everywhere as well. Also move the pressure plate lever while washing.
- If you have the feeling there still is sediment trapped in the base of the device or in the screen cover, you can unscrew both parts to clean them directly.

Tips and Tricks

- **Force Rebooting** : You can force the device to reboot by applying a small magnet on the back side of the device at the level of the display. This will turn the device off.





- **Wheel jamming** : This is of course always a possibility with old/dirty line and the main contributing factor will be the lack or little tension of that line. If you notice the wheel is not spinning first check if the line has enough tension and if possible, adding an extra wrap on a tie off to increase the tension often solves the problem. Otherwise, a way of unblocking the wheel is to hold the line in a hand, increase the tension and then swiftly roll back and forth the mnemo to clean the wheel, that should do the trick probably in 90% of the cases. Of course, after that you'll have to resurvey that shot. From v3.4.2, BASIC and ECO let you check that the wheel turns before you start a shot: see Is the wheel on the line?.
- **Obstacles on the line** :It is important to know that the device measures distance whether you move it forward or backward. This is done intentionally. In case you have an obstacle on the line (thick knot, arrow etc ..) , roll the mnemo all the way to the obstacle, then back the length of the obstacle plus the size of the Mnemo. Unclip it from the line (do not stop the survey or add a station) and clip it on again after the obstacle. Resume rolling the device until you reached the next station.Like this there is no loss of time to rearrange the line.
- **Compensation for thick tie wraps**: You can compensate the loss in measurement of the round part of a large tie wrap by rolling back the device the estimated distance that is missing.
- **Display or not display** :We recommend all user of Mnemo to get familiar enough with the device to use only the colors of the screen and not read the menu at each station. This will increase the smoothness of the process and allows them to focus on stabilizing and clipping on and off the device.
- **Opening the pressure plate (gate)** : If you have difficulty manipulating the pressure plate, brute force won'T help, simply hold the lever between thumb and index, apply a rotational force forward and THEN pull backward.
- **Organizing work, multiple sections** : I personally call all my section AA1 or use BASIC mode. In most cases I can remember what I survey and in which order. In more complex

scenario you can either take the time to change the name of the section or write on a slate complementary information.

- **Trapped water after a dive** : When coming out of a dive, droplets of water can stay trapped for a few hours below the screen cover, where the contrast sensors that read the command button are located. When this happens, the device can no longer properly interpret the sensor values and will stop responding. To resolve this temporary issue, either immerse the device in water again or shake it vigorously to remove the remaining water.

Report a Hardware or Software Issue

If your Mnemo is not behaving as expected — whether it is a hardware failure, a firmware bug, or an unexpected behaviour during a survey — please fill in the failure report form and send it to us.

The report helps us identify patterns, improve the device, and assist you as quickly as possible.

What to report

- Device not powering on or off correctly
- Screen issues (missing pixels, flickering, blank display)
- Compass errors or calibration failures
- Incorrect depth or distance readings
- Firmware crashes or unexpected resets
- Connectivity problems (Wi-Fi, USB)
- Waterproofing concerns or physical damage
- Any behaviour that differs from what the manual describes

How to submit

1. Open the report form using the button below
2. Print it or save it as a PDF (`Ctrl+P` → *Save as PDF*)
3. Fill it in — the more detail the better
4. Send it to sebastien@arianesline.com or open an issue on GitHub

Open Failure Report Form

Tips for a useful report

- Note your **firmware version** before submitting (visible in the main menu)
- Include the **dive number and conditions** when the issue occurred
- Attach a **photo or video** if the failure is visible
- If the device crashed, describe the **exact sequence of button presses** that led to it